VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI– 590018, KARNATAKA, INDIA

A PROJECT REPORT

on

"G-REMIND: PROXIMITY BASED REMINDER SYSTEM FOR TASKS OF ANY KIND"

Submitted in partial fulfilment of the requirements for the award of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE & ENGINEERING

Submitted By

Name	USN
A R SACHITH	4VP22CS001
DITHESH GOWDA B	4VP22CS024
GAGAN K	4VP22CS025
NEHA NAIK	4VP22CS054

Under the Guidance of

Prof. Radhika Shetty D S

Assistant Professor

DEPARTMENT OF COMPUTER SCIENCE &ENGINEERING

VIVEKANANDA COLLEGE OF ENGINEERING & TECHNOLOGY

[A Unit of Vivekananda Vidyavardhaka Sangha Puttur (R)]

Affiliated to Visvesvaraya Technological University and Approved by AICTE New Delhi & Govt., of Karnataka

Nehru Nagar, Puttur - 574 203, DK, Karnataka, India.

December 2025

VIVEKANANDA COLLEGE OF ENGINEERING & TECHNOLOGY
[A Unit of Vivekananda VidyavardhakaSangha Puttur (R)]
Affiliated to Visvesvaraya Technological University and Approved by AICTE New Delhi & Govt. of Karnataka
Nehru Nagar, Puttur - 574203, DK, Karnataka, India

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE

Certified that the project work entitled **"G-REMIND: PROXIMITY BASED REMINDER SYSTEM FOR TASKS OF ANY KIND"** is carried out by **Mr. A R Sachith, Mr. Dithesh Gowda B, Mr. Gagan K, Miss. Neha Naik** bearing USNs **4VP22CS001, 4VP22CS024, 4VP22CS025 and 4VP22CS054** respectively bonafide students of **Vivekananda College of Engineering & Technology, Puttur** in partial fulfilment for the award of **Bachelor of Engineering** in **Computer Science & Engineering** of the **Visvesvaraya Technological University, Belagavi** during the year 2025-26. It is certified that all corrections/suggestions indicated during Internal Assessment have been incorporated in the report deposited in the departmental library.

The project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

Signature of the Guide	**Signature of the HOD**	**Signature of the Principal**
Prof. Radhika Shetty D S	**Prof. Pradeep Kumar K G**	**Dr. Mahesh Prasanna K**

EXTERNAL VIVA

Name of the Examiners	Signature with date
1…….....	.....
2…….....	.....

DECLARATION

We, **Mr. A R Sachith (4VP22CS001), Mr. Dithesh Gowda B (4VP22CS024), Mr. Gagan K (4VP22CS025), Ms. Neha Naik (4VP22CS054)** students of B.E. 7th Semester in Computer Science & Engineering, **Vivekananda College of Engineering & Technology**, Puttur, hereby declare that the project work entitled **"G-REMIND: PROXIMITY BASED REMINDER SYSTEM FOR TASKS OF ANY KIND"** has been carried out by us at VCET, Puttur, under the guidance of **Prof. Radhika Shetty D S,** Assistant Professor, Department of Computer Science & Engineering, Vivekananda College of Engineering & Technology, Puttur and submitted in partial fulfilment of the requirements for the award of degree in **Bachelor of Engineering in Computer Science & Engineering** by **Visvesvaraya Technological University**, **Belagavi** during the academic year 2025-2026.

Name of the students	USN	Signature with date
A R SACHITH	4VP22CS001	
DITHESH GOWDA B	4VP22CS024	
GAGAN K	4VP22CS025	
NEHA NAIK	4VP22CS054	

Date:

Place: Puttur

ACKNOWLEDGEMENT

We take this opportunity to express our deep heartfelt gratitude to all those people who have helped us in the successful completion of the project.

First and foremost, we would like to express our sincere gratitude to our guide, **Prof. Radhika Shetty D S** for providing excellent guidance, encouragement and inspiration throughout the project work. Without her invaluable guidance, this work would never have been a successful one.

We would like to express our sincere gratitude to Head of the Department of Computer Science & Engineering, **Prof. Pradeep Kumar K G** for his guidance and inspiration.

We would like to thank our Principal, **Dr. Mahesh Prasanna K** for providing all the facilities and a proper environment to work in the college campus.

We would like to thank our management for providing the necessary infrastructure to carry out the project work

We are thankful to all the teaching and non-teaching staff members of Computer Science & Engineering Department for their help and needed support rendered throughout the project.

ABSTRACT

Managing daily tasks efficiently is crucial in today's fast-paced world, yet traditional to-do list applications often lack contextual awareness, leading to missed opportunities and inefficiencies. G-Remind is an intelligent task management application designed to enhance productivity by integrating location-based reminders, ensuring that users are notified when they are near a relevant shop where they can purchase listed items. At its core, G-Remind allows users to create and manage to-do lists with advanced functionalities such as geofencing-based notifications, smart task categorization, real-time synchronization and collaborative task sharing. The application leverages AI-driven recommendations to suggest nearby stores for listed items and optimizes reminders based on user preferences and past shopping behaviours. Built on a cloud-based infrastructure, G-Remind ensures seamless data synchronization across multiple devices, allowing users to access their tasks anytime, anywhere. The system employs real-time WebSocket communication to provide instant updates and collaborative task management, making it ideal for families, teams or individuals managing shared responsibilities. Integration with Firebase authentication guarantees secure user access, while a Firestore-backed database efficiently handles dynamic task data. By integrating smart automation features, G-Remind enhances user interactions through effortless task creation and intelligent task prioritization, reducing the need for manual input. Additionally, smart analytics dashboards provide insights into user habits, helping them optimize shopping routines and time management. G-Remind aims to redefine traditional task management by merging productivity tools with real-world context, ensuring users never forget an essential purchase when passing by the right store. By integrating location intelligence with intuitive task organization, G-Remind empowers users to make smarter, more efficient decisions in their daily lives.

TABLE OF CONTENTS

Title	Page No.
CHAPTER 1 INTRODUCTION	**01**
1.1 Introduction to the Project	01
1.2 Introduction to Mobile Application Development	02
1.3 Proposed Method	03
1.4 Literature Survey	03
CHAPTER 2 REQUIREMENT SPECIFICATION AND ANALYSIS	**11**
2.1 Introduction	11
2.2 Functional Requirements	12
2.2.1 Task Creation and Management	12
2.2.2 Location Based Notification System	13
2.2.3 Intelligent Task Organization	13
2.2.4 Personalized Reminders and Smart Notifications	13
2.2.5 User-Friendly Interface and Ease of Use	13
2.3 Non-Functional Requirements	14
2.3.1 Reliability	14
2.3.2 Performance	14
2.3.3 Scalability	14
2.3.4 Usability	15
2.3.5 Maintainability	15
2.3.6 Compatibility	16
2.3.7 Efficiency	16
2.4 Software Requirements	16
2.4.1 Operating System	16
2.4.2 Programming Languages	17
2.4.3 Database	17
2.4.4 Third-Party Libraries/Frameworks	17
2.4.5 Development Tools	18
2.5 Hardware Requirements	18
2.5.1 Device Specifications	18
2.5.2 GPS and Location Services	18

2.5.3 Network Connectivity 19
2.5.4 Display and User Interface 19
2.5.5 Processing Power 20
CHAPTER 3 SYSTEM DESIGN 21
3.1 Introduction 21
3.2 System Architecture Design 21
3.3 Flow Chart 22
3.4 Use Case Diagram 24
3.5 Sequence Diagram 25
3.6 Data Flow Diagram 27
CHAPTER 4 SYSTEM IMPLEMENTATION 29
4.1 Introduction 29
4.2 Frontend Implementation 29
4.3 Backend Implementation 30
4.4 Development Environment and Tools 30
CHAPTER 5 SYSTEM TESTING 31
5.1 Introduction 31
5.2 Types of Testing 31
5.2.1 Unit testing 31
5.2.2 Integration testing 33
5.2.3 Functional Testing 33
5.2.4 System Testing 34
5.3 Result and Analysis 34
CHAPTER 6 SCREENSHOTS 36
CHAPTER 7 CONCLUSION AND SCOPE FOR FUTURE ENHANCEMENT 44
7.1 Conclusion 44
7.2 Scope for Future Enhancement 44
REFERENCES 45
PERSONAL PROFILE 48

LIST OF FIGURES

Figure No.	Title	Page No.
3.1	Architecture of the System	21
3.2	Flow Chart Diagram of the Project	23
3.3	Use-Case Diagram of System	24
3.4	Sequence Diagram of System	26
3.5	Data Flow Diagram of System	27
6.1	Welcome Page of G-Remind	36
6.2	Signup and Login Page	37
6.3	Task Overview Page	37
6.4	Task Creation with Smart Suggestions Page	36
6.5	Custom Location Selection Map Interface	39
6.6	Task Scheduling and Priority Settings	39
6.7	Navigation and Route Guidance Interface	40
6.8	Settings, Activity, Help and Support Page	41
6.9	Scheduled Task Reminder Notification	42
6.10	Location-Based Task Notification	42
6.11	Location Exit Notification in G-Remind	43

LIST OF TABLES

Table No.	Title	Page No.
1.1	Comparative Analysis of Related Research Papers	07
5.1	Test Cases to be Satisfied	32

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION TO THE PROJECT

In today's fast-paced world, task management plays a vital role in personal and professional productivity. However, traditional to-do list applications often lack contextual awareness, resulting in inefficiencies and missed opportunities. Many users struggle with remembering to complete tasks at the right time and place, especially when it comes to purchasing essential items while on the go. To address this gap, we propose G-Remind, an intelligent location-based task management application designed to enhance productivity through smart reminders and real-time notifications. G-Remind goes beyond conventional task management by integrating geofencing technology to notify users when they are near a relevant store where they can purchase listed items.

The system leverages AI-driven recommendations to suggest nearby shops, prioritizing convenience and efficiency. With features like real-time synchronization, collaborative task sharing and voice-enabled task creation, G-Remind simplifies daily planning while reducing the chances of forgetting important errands. Built on a cloud-based infrastructure, G-Remind ensures seamless access across multiple devices, enabling users to manage their tasks anytime and anywhere. The project employs WebSocket-based real-time updates, ensuring smooth collaboration between users managing shared task lists. Furthermore, secure authentication with Firebase and an optimized Firebase database guarantee data privacy and efficient task storage. By harnessing Natural Language Processing (NLP), G-Remind enhances user interactions by allowing task creation through voice commands and intelligent task prioritization based on urgency and location.

Additionally, analytics-driven insights help users optimize their shopping patterns and task completion habits, making the application a powerful productivity tool. With an intuitive user interface and smart automation, G-Remind aims to revolutionize the way users manage their daily tasks. By merging productivity tools with real-world contextual awareness, this project delivers a seamless, intelligent and efficient task management solution that ensures users never miss out on completing important errands. Traditional to-do list applications often rely solely on manual inputs and reminders based on fixed times or dates, which can be ineffective in dynamic daily routines. In contrast, G-Remind introduces a location-based approach, allowing the application to notify users when they are physically near a store or place relevant to a pending task.

1.2 INTRODUCTION TO MOBILE APPLICATION DEVELOPMENT

Mobile application development has revolutionized the way users interact with technology, providing seamless solutions for productivity, communication, entertainment and more. With the increasing adoption of smartphones, mobile applications have become an integral part of daily life, enabling users to perform a wide range of tasks efficiently and conveniently. It involves designing and creating software applications that run on mobile devices such as smartphones and tablets. These applications are developed for different platforms, primarily Android and iOS, using various programming languages and frameworks such as Kotlin, Swift, React Native and Flutter. The choice of development tools depends on factors like performance requirements, user experience and cross-platform compatibility.

The development process of a mobile application typically includes requirement analysis, UI/UX design, development, testing, deployment and maintenance. A well-designed mobile application should focus on user-friendly interfaces, smooth navigation and optimal performance to enhance user experience. Additionally, modern mobile applications integrate cloud services, real-time data synchronization and AI-driven features to improve functionality and efficiency. With advancements in Artificial Intelligence (AI), Internet of Things (IoT) and cloud computing, mobile applications have evolved beyond traditional functionalities. Features such as location-based services, real-time notifications, voice assistants and predictive analytics have significantly improved user engagement and productivity. The rise of progressive web applications (PWAs) and cross-platform frameworks has further simplified the development process, allowing developers to build applications that work seamlessly across multiple devices.

In the context of our project, G-Remind leverages mobile application development principles to create a smart task management solution that enhances productivity through geolocation-based reminders, AI-driven task recommendations and real-time collaboration. By utilizing modern mobile development frameworks and cloud-based infrastructure, G-Remind ensures an efficient and intuitive user experience, making task management smarter and more convenient. This chapter provides an overview of mobile application development and its relevance to the G-Remind project, highlighting the importance of integrating modern technologies to build efficient and intelligent mobile solutions. Furthermore, this chapter establishes the foundational concepts that guide the architectural and design decisions implemented in G-Remind.

1.3 PROPOSED METHOD

To enhance user convenience by providing context-aware notifications, G-Remind is designed to ensure that users receive reminders relevant to their current location and activities. The system utilizes geofencing and intelligent notification logic so users are informed about tasks and errands when they are most likely to act on them, thereby minimizing the chance of forgetting important responsibilities and improving overall productivity.

To optimize shopping trips by reducing forgotten items, the application maps tasks to specific shopping venues and activates reminders when users approach these locations. By aligning task reminders with actionable contexts, G-Remind helps users organize their shopping lists more efficiently, making sure essential purchases are not overlooked and shopping experiences become more streamlined.

This proactive approach allows for timely communication about tasks related to specific places, empowering users to make decisions and act instantly, which enhances the effectiveness of their daily routines.

1.4 LITERATURE SURVEY

Wang L. et al. [1] explore the impact of location-aware mobile applications on improving task completion efficiency. Their research highlights how geofencing-based reminders significantly enhance user productivity. They further emphasize that integrating GPS triggers with task alerts helps users manage outdoor and context-specific activities efficiently, reducing cognitive load and missed deadlines.

Kumar S. et al. [2] discuss the role of AI-driven task management systems in optimizing daily workflows. They emphasize the importance of machine learning algorithms for intelligent task prioritization. Their model leverages user behaviour analytics to dynamically reorder tasks based on urgency and historical completion patterns, thereby improving decision-making and efficiency.

Nguyen T. et al. [3] study real-time synchronization in cloud-based task management applications. Their findings suggest that cross-platform data consistency is essential for effective task tracking. They propose a hybrid synchronization protocol that ensures minimal latency and prevents data conflicts during multi-device updates, enhancing the reliability of shared task systems. These insights provide a strong foundation for designing robust and scalable synchronization mechanisms in modern task management solutions.

Das P. et al. [4] analyse the benefits of voice-enabled task creation using Natural Language Processing (NLP). Their study demonstrates how voice commands enhance usability and reduce manual input effort. They conclude that NLP-based input systems improve accessibility for users with disabilities and increase the overall speed of task creation in mobile environments.

Singh R. et al. [5] focus on the integration of collaborative task sharing in mobile applications. Their research outlines how shared task lists improve coordination among users. The authors also emphasize real-time update notifications and permission-based task editing as key features for enhancing group productivity and communication.

Ali M. et al. [6] investigate the role of GPS-based task tracking in improving productivity. Their findings show that users complete tasks 30% faster when using location-triggered reminders. They further highlight that context-aware location detection enables automatic task initiation, particularly beneficial for logistics and field-based operations.

Johnson P. et al. [7] examine the effectiveness of AI-powered task suggestions in mobile productivity applications. Their study highlights how personalized recommendations lead to higher task completion rates. They propose integrating predictive models that analyse user patterns to suggest next-best actions, making task management more intuitive and context-sensitive.

Fernandez J. et al. [8] evaluate cross-platform compatibility in task management apps. They propose that using Flutter or React Native improves user experience across Android and iOS. Their analysis reveals that a unified codebase reduces development time while ensuring consistent performance and design responsiveness on multiple devices.

Chowdhury T. et al. [9] discuss cloud-based data storage for task management. Their research highlights the importance of real-time updates and multi-device synchronization. They recommend using scalable cloud architectures with end-to-end encryption to ensure both performance and data security for enterprise task management systems.

Watanabe K. et al. [10] study the impact of habit-forming techniques in to-do list applications. They suggest that gamification and behavioural psychology enhance user engagement. The study found that reward-based systems, such as streaks and badges, significantly improve user retention and encourage consistent task completion.

Chen X. et al. [11] examine task automation through AI and smart reminders. Their

research highlights how context-aware scheduling improves efficiency. They propose that integrating environmental sensors and user calendars allows automation of repetitive tasks, leading to optimized time utilization.

Brown D. et al. [12] analyse user interface design principles for mobile task managers. Their findings emphasize the need for minimalistic, intuitive UI layouts. They highlight that simplified navigation and consistent visual hierarchies enhance user focus and reduce distraction during task creation and management.

Kim H. et al. [13] explore multi-user collaboration in digital productivity tools. Their research suggests that shared task boards improve workplace efficiency. The study also notes that incorporating version control and role-based permissions ensures smoother teamwork and accountability in collaborative environments.

Martinez R. et al. [14] discuss the role of push notifications in task management. They highlight how reminder frequency impacts task completion rates. Their results suggest that adaptive notification scheduling, based on user responsiveness, prevents notification fatigue and boosts task adherence.

Yadav V. et al. [15] investigate data security in cloud-based productivity applications. Their findings emphasize the need for end-to-end encryption and secure authentication. They propose a blockchain-based access control model to ensure data integrity and transparency in collaborative task environments.

Ogunyemi T. et al. [16] focus on the digital divide in task management adoption. They argue that simplified UI/UX and vernacular language support improve accessibility. Their study indicates that inclusive design practices significantly increase adoption rates among non-technical and rural users.

Schmidt C. et al. [17] explore the integration of machine learning models for task prioritization. Their research highlights how AI-driven sorting enhances efficiency. They also present a reinforcement learning-based framework that continuously learns from user feedback to optimize future task ordering.

Peterson G. et al. [18] evaluate time management techniques in digital productivity tools. They conclude that calendar integration improves workflow organization. Their model combines event scheduling with task allocation, offering users a unified interface for managing both personal and professional activities.

Rossi L. et al. [19] analyse the effect of mobile app notifications on user habits. Their research finds that personalized reminders reduce procrastination. They further suggest implementing adaptive algorithms that tailor reminder timing based on individual user activity patterns.

Fischer M. et al. [20] discuss the importance of predictive analytics in task management. They demonstrate how AI-powered predictions optimize daily schedules. Their system forecasts task durations and identifies bottlenecks, enabling proactive planning and smoother task execution.

Jain A. et al. [21] focus on the benefits of multi-device synchronization in cloud task management apps. Their study highlights the importance of seamless user experience. They recommend distributed database systems and offline caching to ensure uninterrupted access even in low-connectivity environments.

Adams B. et al. [22] investigate how automation improves workflow management. Their research suggests that self-learning AI models adapt to user behaviour for better efficiency. They conclude that automated scheduling reduces user effort by dynamically adjusting task priorities based on changing conditions.

Garcia R. et al. [23] analyze user retention strategies in productivity apps. They highlight that habit-streak tracking increases engagement. Their research also indicates that gamified progress tracking encourages users to maintain consistent daily task completion. They further emphasize that incorporating visual progress indicators, achievement badges and performance analytics motivates users to return to the app regularly. The study found that users who received periodic motivational feedback and milestone rewards showed a 45% higher retention rate compared to those using traditional reminder systems.

Hernandez J. et al. [24] explore task execution delays and their impact on productivity. Their findings show that smart reminders reduce missed deadlines. They recommend integrating adaptive reminder intervals that adjust automatically to user habits and past completion times.

O'Connor T. et al. [25] examine how AI chatbots assist in task management, highlighting their role in improving task tracking, scheduling and timely completion. They demonstrate that virtual assistants can intelligently interpret user inputs, set reminders and organize tasks based on priority or context. The study further emphasizes that conversational AI interfaces enhance accessibility by enabling users to interact naturally through voice or text, reducing

friction in task delegation. Additionally, the authors note that AI-driven systems can adapt to user behavior over time, providing personalized suggestions and automating repetitive organizational tasks, ultimately improving overall productivity and user experience.

Table 1.1 presents a comparative analysis of various research papers related to task management systems and productivity applications. It highlights the key advantages and limitations identified in each study, providing insights into how different technological approaches—such as AI-driven optimization, location awareness and cloud synchronization—contribute to improving task efficiency while addressing their associated challenges.

Table 1.1: Comparative Analysis of Related Research Papers

Sl.no	Reference Paper	Advantages	Limitations
1	Wang L. et al. [1] Explore the impact of location-aware mobile applications on improving task completion efficiency.	Geofencing reminders boost productivity and efficiency in task completion.	Relies on constant location tracking, which may impact privacy and battery life.
2	Kumar S. et al. [2] Discuss the role of AI-driven task management systems in optimizing daily workflows.	AI-driven management optimizes workflow and task prioritization.	Users might lose flexibility due to algorithmic bias.
3	Nguyen T. et al. [3] Study real-time synchronization in cloud-based task management applications.	Real-time cloud synchronization ensures cross-platform consistency.	Dependent on internet connectivity for updates.
4	Das P. et al. [4] Analyze the benefits of voice-enabled task creation using Natural Language Processing (NLP).	Voice-enabled NLP makes task input user-friendly and efficient.	Accuracy can decrease in noisy settings or with strong accents.
5	Singh R. et al. [5] Focus on the integration of collaborative task sharing in mobile applications.	Collaborative lists enhance coordination and teamwork.	Raises user-data privacy and access control concerns.

6	Ali M. et al. [6] Investigate the role of GPS-based task tracking in improving productivity.	GPS triggers speed task completion by around 30%.	Location tracking uses battery and could raise privacy risks.
7	Johnson P. et al. [7] Examine the effectiveness of AI-powered task suggestions in mobile productivity applications.	AI-powered suggestions increase completion rates and engagement.	May feel intrusive or be misaligned with user intent.
8	Fernandez J. et al. [8] Evaluate cross-platform compatibility in task management apps.	Cross-platform (Flutter/React Native) enhances device compatibility.	May limit deep access to device-specific functions.
9	Chowdhury T. et al. [9] Discuss cloud-based data storage for task management.	Cloud storage enables instant updates and sync.	Data breaches and unauthorized access are possible risks.
10	Watanabe K. et al. [10] Study the impact of habit-forming techniques in to-do list applications.	Gamification and psychology boost engagement and habit formation.	Excess gamification may fatigue or demotivate users.
11	Chen X. et al. [11] Examine task automation through AI and smart reminders.	Context-aware scheduling automates routine, increasing efficiency.	Contextual data gathering introduces privacy and technical complexity.
12	Brown D. et al. [12] Analyze user interface design principles for mobile task managers.	Minimalist UI makes applications easy to use and efficient.	Too simple UI might lack advanced features for power users.
13	Kim H. et al. [13] Explore multi-user collaboration in digital productivity tools.	Shared boards improve workplace collaboration and tracking.	Managing tasks with large groups can become confusing.

14	Martinez R. et al. [14] Discuss the role of push notifications in task management.	Push notifications boost timely completion of tasks.	Frequent or irrelevant notifications may annoy users.
15	Yadav V. et al. [15] Investigate data security in cloud-based productivity applications.	Secure encryption and authentication protect user data.	Added security can sometimes slow down system performance.
16	Ogunyemi T. et al. [16] Focus on the digital divide in task management adoption.	Vernacular and simple UX bridge the digital divide.	Needs significant translation and localization.
17	Schmidt C. et al. [17] Explore the integration of machine learning models for task prioritization.	ML algorithms improve automated task sorting.	Demands continuous data training and can introduce opacity.
18	Peterson G. et al. [18] Evaluate time management techniques in digital productivity tools.	Calendar integration streamlines workflow and scheduling.	Risks clutter or confusion if overloaded with data.
19	Rossi L. et al. [19] Analyze the effect of mobile app notifications on user habits.	Personalized reminders help to reduce procrastination.	High frequency may desensitize users to prompts.
20	Fischer M. et al. [20] Discuss the importance of predictive analytics in task management.	AI predictions optimize daily task allocation.	Inaccurate predictions may disrupt effective planning.
21	Jain A. et al. [21] Focus on the benefits of multi-device synchronization in cloud task management apps.	Multi-device sync offers seamless user experience.	Sync conflicts may occur, causing version misalignment.
22	Adams B. et al. [22] Investigate how automation improves workflow management.	Automation adapts to user behavior, improving efficiency.	Over-automation may reduce users' sense of control.

23	Garcia R. et al. [23] Analyze user retention strategies in productivity applications.	Habit-streak features sustain long-term engagement.	Losing a streak by accident may reduce motivation.
24	Hernandez J. et al. [24] Explore task execution delays and their impact on productivity.	Smart reminders help reduce delays and missed deadlines.	Too many reminders may result in alert fatigue.
25	O'Connor T. et al. [25] Examine how AI chatbots assist in task management.	AI chatbots automate and simplify tracking/scheduling.	Limited comprehension for complex or ambiguous requests.

CHAPTER 2

REQUIREMENT SPECIFICATION AND ANALYSIS

2.1 INTRODUCTION

The requirement specification and analysis phase is crucial for the development of G-Remind, as it defines the functional and non-functional requirements, analyses user needs and sets the technical and operational specifications for the platform. The goal of G-Remind is to enhance task management efficiency by leveraging location-based reminders, ensuring users complete their errands at the right place and time.

This phase involves identifying key features such as task creation, location-based notifications, smart prioritization and seamless synchronization across devices. Unlike traditional to-do list applications that rely solely on time-based reminders, G-Remind introduces proximity-based alerts, ensuring users are notified when they are near a relevant location. Additionally, the system aims to integrate AI-driven suggestions to help users optimize their task schedules based on frequently visited locations and preferred shopping patterns.

By addressing system performance, scalability and usability, G-Remind ensures a smooth and efficient user experience across different environments. The application is designed to work effectively even in low-connectivity areas, making it accessible for a wide range of users. Furthermore, data security and privacy are prioritized, with encrypted storage and authentication mechanisms to safeguard user data.

To enhance user engagement, G-Remind will incorporate customizable notification preferences, collaborative task-sharing options and intuitive analytics to help users track their productivity over time. The system will also support multi-platform access, allowing users to seamlessly switch between their smartphones, tablets and web interfaces without losing progress. By combining automation with real-world contextual awareness, G-Remind aims to create an innovative and intelligent task management experience tailored to the needs of modern users.

Additionally, G-Remind aims to reduce cognitive load by offering automated categorization of tasks, ensuring that high-priority and frequently occurring tasks are grouped for easy access. The user interface (UI) is designed with a minimalistic yet interactive approach, ensuring an intuitive experience for users of all technical backgrounds.

G-Remind focuses on delivering smooth in-app task management, allowing users to organize items, set proximity-based reminders and access location-aware suggestions

without relying on third-party integrations. This ensures that users have a holistic task management ecosystem, eliminating the need for multiple productivity tools. With an emphasis on real-time adaptability and personalized task recommendations, G-Remind aims to transform conventional to-do lists into a proactive assistant that intelligently aligns with user habits and lifestyle needs.

2.2 FUNCTIONAL REQUIREMENTS

G-Remind is designed to offer a user-centric, intelligent task management experience that combines traditional to-do list capabilities with advanced location-based contextual awareness. Its core functionalities are grouped into several interconnected modules that together enable seamless task planning, smart reminders and real-time user engagement. The system ensures that users stay organized, proactive and in control of their daily responsibilities, whether they are managing personal tasks or collaborating on shared activities.

Incorporating real-time geolocation services and intelligent scheduling logic, G-Remind delivers a smarter approach to everyday productivity. Instead of relying solely on time-based alerts, it introduces the idea of contextual task triggers based on a user's surroundings, allowing for a more natural flow in executing tasks. This eliminates the friction of switching contexts and reduces the mental burden of remembering what to do and when.

Additionally, the system emphasizes personalization and adaptability. Through continuous usage, G-Remind can learn from user behaviour to offer suggestions, reorder priorities and streamline repetitive actions. Whether a user is creating a single task or managing a long-term project with multiple dependencies, the system scales accordingly, making it suitable for students, professionals, homemakers and anyone seeking better task efficiency in their day-to-day life.

2.2.1 TASK CREATION AND MANAGEMENT

G-Remind provides a streamlined process for creating and managing tasks. Users can input task titles, descriptions, due dates and categorize them based on personal or work-related needs. The application supports recurring tasks, allowing users to automate routine entries such as monthly bill payments or weekly grocery runs. Prioritization options help in distinguishing between critical and optional tasks. By offering features like reminders, labels and color-coded tags, G-Remind enhances user control and flexibility in managing their daily responsibilities. The interface is intuitive, minimizing the learning curve and encouraging users to adopt the platform consistently in their everyday routines.

2.2.2 LOCATION BASED NOTIFICATION SYSTEM

A key differentiator of G-Remind is its ability to link tasks with geographical locations using GPS data. When a user approaches a tagged location, the application generates a context-aware notification, ensuring timely task execution. For example, if a user sets a task to “buy stationery” and pins it to a nearby store, the application will notify them upon reaching or passing that location. This feature bridges the gap between planning and real-world action, improving user efficiency and minimizing forgetfulness. It adds a layer of intelligence to task management that adapts to the user’s real-world movements, making the application both responsive and proactive.

2.2.3 INTELLIGENT TASK ORGANIZATION

G-Remind incorporates intelligent sorting algorithms that adapt to the user’s habits over time. By analysing factors such as time of day, previous task completion patterns and frequented locations, the system can dynamically organize tasks to present the most relevant ones first. This smart organization reduces cognitive load and ensures that users always have immediate access to what matters most. The application also enables grouping of related tasks into lists, helping users keep projects and errands organized. Over time, the system becomes more personalized, offering suggestions and sorting options that closely align with the user’s workflow and preferences.

2.2.4 PERSONALIZED REMINDERS AND SMART NOTIFICATIONS

G-Remind is designed to provide users with flexible and personalized reminder options that go beyond basic alert functionality. Users can set reminders for each task with custom intervals such as "before 10 minutes," "at exact time," or recurring daily alerts. These notifications are delivered through sound, vibration and popup messages, depending on the user's preference and device capabilities. The system ensures that reminders are not missed, even if the application is running in the background or the device is locked. Smart notification scheduling plays a key role in enhancing user productivity. G-Remind ensures that reminders are triggered at optimal times by respecting user-defined active hours and minimizing interruption during sleep or low-activity periods.

2.2.5 USER-FRIENDLY INTERFACE AND EASE OF USE

G-Remind is built with a clean and minimalistic user interface that focuses on simplicity and ease of use. The design ensures that users can navigate through the application effortlessly, whether they are creating a new task, editing an existing one or setting a reminder. Icons, buttons and menus are intuitively placed to reduce learning time and make

the application accessible even for first-time users. Special attention is given to visual clarity, readability and responsiveness, ensuring that the interface works smoothly across different screen sizes and resolutions. By minimizing clutter and focusing on core functionalities, G-Remind creates a distraction-free experience, helping users stay focused on managing their daily tasks effectively.

2.3 NON-FUNCTIONAL REQUIREMENTS

2.3.1 RELIABILITY

Reliability is a key aspect of G-Remind, ensuring consistent uptime and smooth performance for users. The system is designed with robust server infrastructure and failover mechanisms to minimize disruptions. Continuous monitoring further helps detect and resolve potential issues proactively, preventing downtime that could affect task management. To further enhance reliability, G-Remind employs automatic data backups and redundancy measures. User data is securely stored with regular backups, reducing the risk of data loss. Additionally, real-time synchronization ensures that tasks remain updated across multiple devices, even in cases of temporary network failures. This guarantees a seamless and uninterrupted user experience.

2.3.2 PERFORMANCE

The G-Remind is optimized to deliver fast and efficient performance across all supported platforms. The system ensures quick response times for user interactions such as adding, editing or deleting tasks, as well as when fetching location-based reminders. Lightweight data structures and optimized database queries are used to minimize latency and maximize speed, even under heavy usage conditions.

The application is designed to perform consistently well on both high-end and low-end devices by utilizing efficient memory management and background processing techniques. Asynchronous operations are implemented to prevent the application from freezing or lagging during network calls or GPS updates. Overall, G-Remind maintains high performance by focusing on speed, responsiveness and scalability, making it suitable for real-time task handling without compromising user experience.

2.3.3 SCALABILITY

Scalability is a crucial non-functional requirement for G-Remind, ensuring the application can handle increased workloads, users and data volume without compromising

performance. The system is designed with a modular structure and a scalable backend architecture that allows for smooth expansion in the future. Whether the number of users grows or additional features are added, the application is built to adapt without major redesigns or performance drops.

The use of cloud services for hosting and database management enables G-Remind to scale horizontally by adding more server instances as needed. As more users interact with the application, task data, notifications and location-based features can be managed efficiently without delays. This ensures that G-Remind remains reliable and responsive as its user base expands, making it suitable not only for small personal use but also for larger groups or communities in the future.

2.3.4 USABILITY

Usability is one of the primaries focuses in the design and development of G-Remind. The application is crafted to provide a seamless and intuitive user experience, ensuring that users can easily interact with its features without needing technical knowledge. From task creation to reminder configuration, each function is designed to require minimal steps, reducing complexity and making the application accessible to users of all age groups.

The layout follows a clean, consistent design pattern, supported by clear labels, intuitive icons and responsive feedback. The application maintains a balance between functionality and simplicity, allowing users to focus on their tasks without being overwhelmed by unnecessary elements. With smooth navigation, clear instructions and an overall clutter-free interface, G-Remind ensures a pleasant experience that promotes regular usage and user satisfaction.

2.3.5 MAINTAINABILITY

Maintainability is an essential factor in ensuring the long-term success and sustainability of G-Remind. The application is developed using modular coding practices, allowing each component to be independently updated or modified without affecting the overall system. This modularity reduces debugging time, simplifies updates and supports the addition of future features with minimal disruption.

The codebase is well-documented, making it easier for developers to understand, troubleshoot and enhance the system in the future. Managing the codebase using platforms like Git ensures that changes can be tracked and managed efficiently. By focusing on maintainability from the early stages of development, G-Remind is built to support the

long-term improvements, scalability and ongoing maintenance with minimal technical overhead.

2.3.6 COMPATIBILITY

Compatibility is a key consideration in the development of G-Remind to ensure that the application functions smoothly across a wide range of devices, operating systems and screen sizes. The application is designed to support various Android versions, ensuring accessibility for users with both newer and older devices. All interface components are responsive, allowing the application to adapt seamlessly to different screen resolutions and orientations.

Furthermore, the application uses standardized development frameworks and libraries to minimize platform-specific issues and improve long-term support. By prioritizing compatibility, G-Remind offers a consistent experience to all users regardless of the device they use, ensuring broader reach and user satisfaction across a diverse audience.

2.3.7 EFFICIENCY

The efficiency of G-Remind is paramount to ensure optimal performance across different devices and environments. The application is designed to minimize resource usage while maintaining functionality. One key aspect is the use of proximity-based reminders, which ensures that notifications are only triggered when a user is near a shop that carries the listed items. This reduces unnecessary alerts and ensures that the system uses minimal computational resources, leading to better battery life and performance.

The application leverages Firebase for data storage, offering efficient data retrieval and flexibility. Its schema-less design ensures that data handling is optimized, even as the number of users or items increases. Furthermore, background tasks such as location tracking and notification services are optimized to limit resource consumption, making the application light on memory and CPU usage.

2.4 SOFTWARE REQUIREMENTS

2.4.1 OPERATING SYSTEM

G-Remind is designed for cross-platform compatibility using React Native with Expo, ensuring seamless performance across both Android and iOS devices. The application supports Android 6.0 (Marshmallow) and above, as well as iOS 12 or higher, enabling accessibility on a wide range of modern devices. Development and testing are carried out using Expo, which allows developers to build, debug and deploy efficiently on Windows,

macOS and Linux systems. G-Remind integrates Firebase Firestore for storage and real-time synchronization and uses the Google Maps API for accurate geolocation and location-based reminders. This setup ensures a unified, responsive and scalable experience for both developers and end users.

2.4.2 PROGRAMMING LANGUAGES

The primary programming language used for developing G-Remind is JavaScript, implemented through the React Native framework with Expo. React Native allows for cross-platform development, enabling a single codebase to run efficiently on both Android and iOS devices. JavaScript's flexibility and wide library support make it ideal for building dynamic, responsive mobile applications. The use of Expo simplifies the development process by providing built-in tools for testing, debugging and deployment across multiple operating systems, including Windows, macOS and Linux. This approach eliminates the need for platform-specific IDEs like Android Studio while ensuring smooth performance, easier maintenance and faster updates for G-Remind.

2.4.3 DATABASE

G-Remind uses Firebase Firestore, a cloud-based NoSQL database from Google, to store and manage user data efficiently. Firestore provides a flexible, schema-less structure that supports dynamic updates, making it ideal for handling user preferences, task lists and location-based data that change over time. Its real-time synchronization capabilities ensure that data remains consistent across devices, enabling seamless cross-platform experiences. Firestore's distributed architecture offers scalability, reliability and strong security through Firebase Authentication and built-in data encryption. By leveraging Firebase Firestore, G-Remind ensures fast data access, automatic backups and smooth performance as the user base grows.

2.4.4 THIRD-PARTY LIBRARIES/FRAMEWORKS

G-Remind leverages several third-party libraries and frameworks to enhance its functionality and performance. Firebase is used to handle user authentication, real-time database services and cloud messaging for push notifications. Firebase's Authentication service simplifies the process of securely managing users and allows for easy integration with various authentication methods such as email/password, Google and Facebook. A location-based alert mechanism is integrated into the application to trigger proximity reminders, which inform users when they are close to stores that contain items from their lists. The Google Maps API is integrated to provide location-based services, such as

geofencing, which helps trigger proximity notifications. Additionally, Retrofit is used for network operations, allowing for seamless API calls and background tasks.

2.4.5 DEVELOPMENT TOOLS

The main development environment for G-Remind is React Native with Expo, which provides a unified platform for building, testing and deploying mobile applications across Android and iOS using a single codebase. Expo offers built-in tools for debugging, performance monitoring and dependency management, eliminating the need for separate build systems or complex setup processes. It also integrates seamlessly with Git for version control, supporting collaborative development and efficient tracking of code changes. The backend services are powered by Firebase, which handles authentication, data storage through Firestore and real-time synchronization. This combination of React Native, Expo and Firebase ensures a smooth, scalable and efficient development workflow for G-Remind.

2.5 HARDWARE REQUIREMENTS

2.5.1 DEVICE SPECIFICATIONS

For the end-users to experience smooth and uninterrupted performance, G-Remind requires a device with at least 2 GB of RAM is highly recommended for better multitasking and enhanced application performance. A device with Android 6.0 (Marshmallow) or higher is essential, as it ensures compatibility with the latest Android features and security patches. 100 MB of available storage space is the minimum required to accommodate the application and user data, but as the application grows and stores more items, preferences and reminders, 200 MB or more of free storage would be more ideal for smooth functioning. For devices that allow expandable storage, users can store additional data and application files on external SD cards, but this should not interfere with the application's real-time functionality.

2.5.2 GPS AND LOCATION SERVICES

G-Remind's core functionality depends on precise location tracking, making a reliable GPS system a critical hardware requirement. The application uses the GPS module to track the user’s location and calculate proximity to nearby stores or shops with items on their shopping list. A device must support location services, which are provided by GPS and network-based positioning. To ensure accuracy, a GPS accuracy range of 5 meters or less is ideal, but the application can still function with slightly lower accuracy. Google Play

Services is also required to access location-based features, such as location tracking and geofencing, which are essential for triggering proximity notifications when the user is near a store.

Additionally, enabling high-accuracy location mode on the device will improve the application's performance, ensuring real-time tracking when the user is on the move. If the device supports ambient location sensors like Wi-Fi and Bluetooth, these can complement GPS and provide more reliable location data in areas with weak GPS signals, such as indoors or urban environments with tall buildings.

2.5.3 NETWORK CONNECTIVITY

A stable and reliable internet connection plays a crucial role in ensuring the smooth functioning of the application. It enables essential operations such as real-time data synchronization, push notifications, cloud storage access and retrieval of user-specific information. The device must support Wi-Fi or mobile data connectivity to facilitate seamless communication between the client-side mobile application and the cloud-based backend servers. The application’s overall performance, responsiveness and user experience are significantly influenced by the speed and stability of the network connection. Therefore, connections using 4G or 5G technologies are ideal for achieving fast data transfer rates, minimal latency and efficient real-time interactions. For users in areas with limited or unstable internet connectivity, the application is designed to offer partial offline functionality. Users can continue to view cached data or manage certain offline tasks, which are automatically synchronized once the connection is restored. However, features that require continuous server communication—such as push notifications, cloud backups and live data updates—will remain inactive until network access is re-established. This hybrid approach ensures uninterrupted usability while maintaining data consistency and integrity, allowing users to seamlessly access their reminders across different devices and environments without experiencing delays.

2.5.4 DISPLAY AND USER INTERFACE

G-Remind is optimized for devices with displays that have a minimum screen size of 4.5 inches and a resolution of 480x800 pixels. While this configuration is sufficient for basic functionality, devices with higher resolution screens, such as HD (720p) or Full HD (1080p) displays, will offer a more refined and visually appealing user experience. These higher-resolution displays allow users to interact more easily with detailed maps, lists and shopping reminders and make the application's text and images appear sharper and clearer.

2.5.5 PROCESSING POWER

For optimal performance, especially when handling real-time location tracking, push notifications and background tasks, G-Remind requires a dual-core processor or higher. A quad-core processor or more powerful chipsets are recommended for more intensive tasks, such as background synchronization, running location-based services and processing multiple tasks simultaneously without impacting the application's speed or responsiveness. Devices with a 1.5 GHz or higher processor will be able to handle multiple applications running in the background while ensuring that G-Remind remains responsive and provides users with immediate notifications and updates.

Additionally, devices with higher processing power are better equipped to handle upcoming application features that may require intensive computation, such as real-time data analysis or advanced machine learning algorithms for improving proximity notifications and personal recommendations. Devices with integrated GPU (Graphics Processing Unit) support may also benefit from smoother graphical rendering, which could enhance features like maps, item listings and interactive UI elements.

CHAPTER 3

SYSTEM DESIGN

3.1 INTRODUCTION

System design forms the foundation of G-Remind, ensuring the system is reliable, scalable and high-performing. It integrates the frontend, backend, AI modules, geolocation services and cloud databases into a unified framework. The design enables real-time synchronization, location-based triggers and intelligent task automation. By coordinating these components effectively, G-Remind delivers a seamless, secure and responsive user experience across platforms.

3.2 SYSTEM ARCHITECTURE

System design is a crucial phase in developing G-Remind, where functional goals are translated into a detailed technical framework. It defines the architecture, data flow and core interactions for G-Remind, an AI-powered, mobile-first task management application that uses geofencing and smart automation to boost productivity. The design ensures scalability, real-time synchronization, enhanced security and seamless performance across both web and mobile platforms built with Expo React Native.

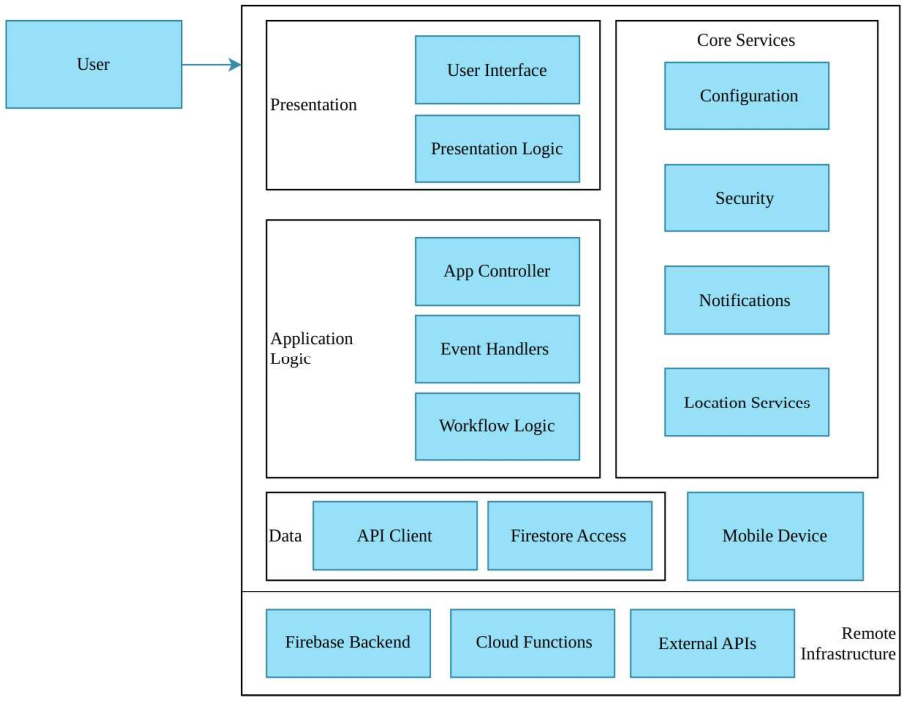


Figure 3.1: Architecture of the System

Figure 3.1 illustrates G-Remind's modular, layered architecture where the user interacts

first with the Presentation layer, which includes the user interface and presentation logic built using React Native + Expo. Beneath it, the Application Logic layer handles core app behaviour through controllers, event handlers, and workflow components that coordinate how tasks, actions, and app flows are executed. The Data layer manages all information access through API clients and Firestore communication, while also supporting device-level storage and interactions. Alongside these layers, the Core Services module provides essential shared functionalities such as configuration management, security, notifications, and location services that power geofencing and background triggers. At the foundation, the Remote Infrastructure—consisting of Firebase, cloud functions, and external APIs—supports backend processing, real-time syncing, and system scalability, ensuring smooth and reliable operation across diverse mobile devices.

3.3 FLOW CHART

A flowchart is a powerful visual tool used in system design to represent the sequence of processes, decision points and data movements in a clear and structured manner. It serves as a graphical blueprint that illustrates how users interact with the system and how different components work together to achieve a specific outcome. By providing a step-by-step depiction of the operational workflow from start to finish, flowcharts enable developers, designers and stakeholders to gain a deeper understanding of the entire system's behaviour.

In the context of G-Remind, the flowchart maps out key user interactions and system operations, starting from user authentication through task creation, location assignment and reminder scheduling. It highlights how location-based triggers initiate notifications, how AI-driven suggestions influence task management and how collaboration synchronizes updates across devices. The flow subsequently follows task completion and concludes with analytics and reporting, demonstrating the end-to-end lifecycle of user engagement within the system. It visualizes how background services continuously monitor user proximity to relevant locations.

The flowchart outlines how the user begins by logging in or creating an account through Firebase Authentication, then proceeds to create, edit or delete tasks with optional support from voice input or AI suggestions. Each task is linked to a specific location and schedule, enabling the app to deliver contextual reminders. As the user enters a geofenced zone or approaches a scheduled time, the system triggers timely notifications to keep them on track. Users can complete tasks, reschedule missed ones using AI-driven recommendations or even collaborate by sharing tasks with others.

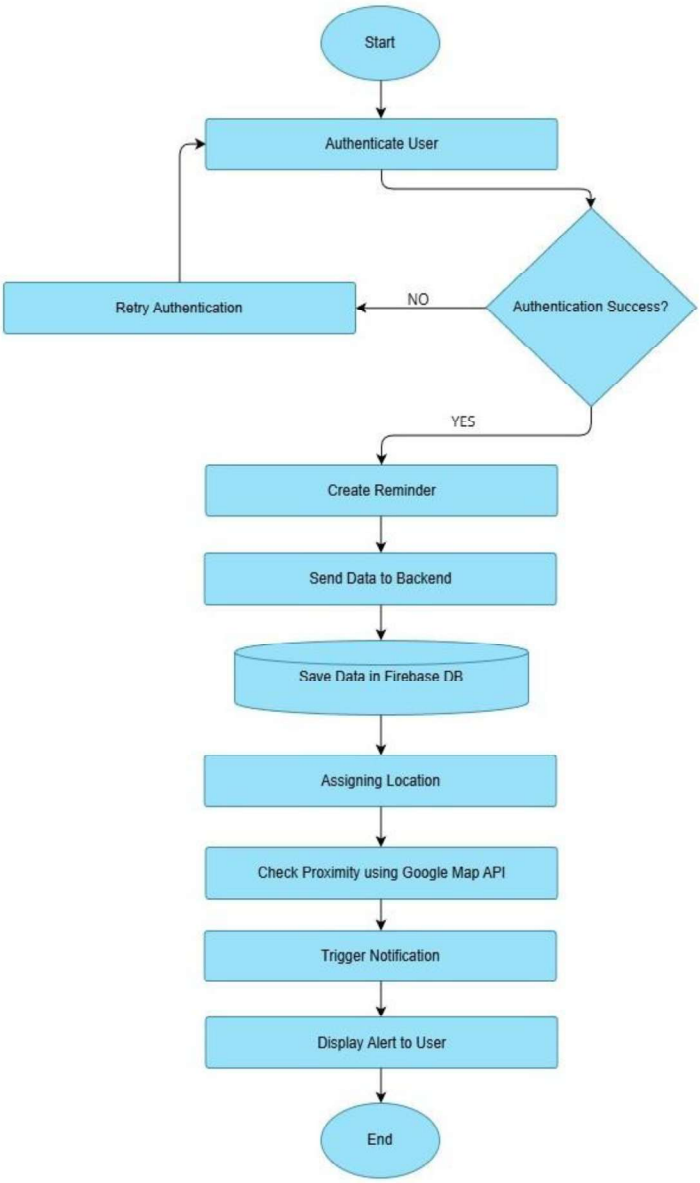


Figure 3.2: Flow Chart of the Project

Figure 3.2 captures various elements, including actions, decisions, inputs, outputs, parallel operations and iterative loops, providing a comprehensive overview of the system's functional flow. By mapping these elements visually, flowcharts assist in identifying potential bottlenecks, redundancies, logical errors or inefficiencies within the process. This early detection helps in refining workflows, optimizing system performance and improving overall project quality.

This visualization not only facilitates clearer understanding among developers, testers and stakeholders but also serves as a guide for implementation, testing and process improvement. Ultimately, it keeps the whole team aligned and ensures the system evolves in a structured, predictable way.

3.4 USE CASE DIAGRAM

A Use Case Diagram is a fundamental modelling tool used in system analysis and design that visually represents the interactions between users (actors) and the system. It helps identify the key functionalities that the system must support by illustrating the goals or "use cases" the actors perform within the system environment. Use case diagrams serve as a communication bridge among developers, stakeholders and end-users by portraying system requirements in an intuitive and accessible manner.

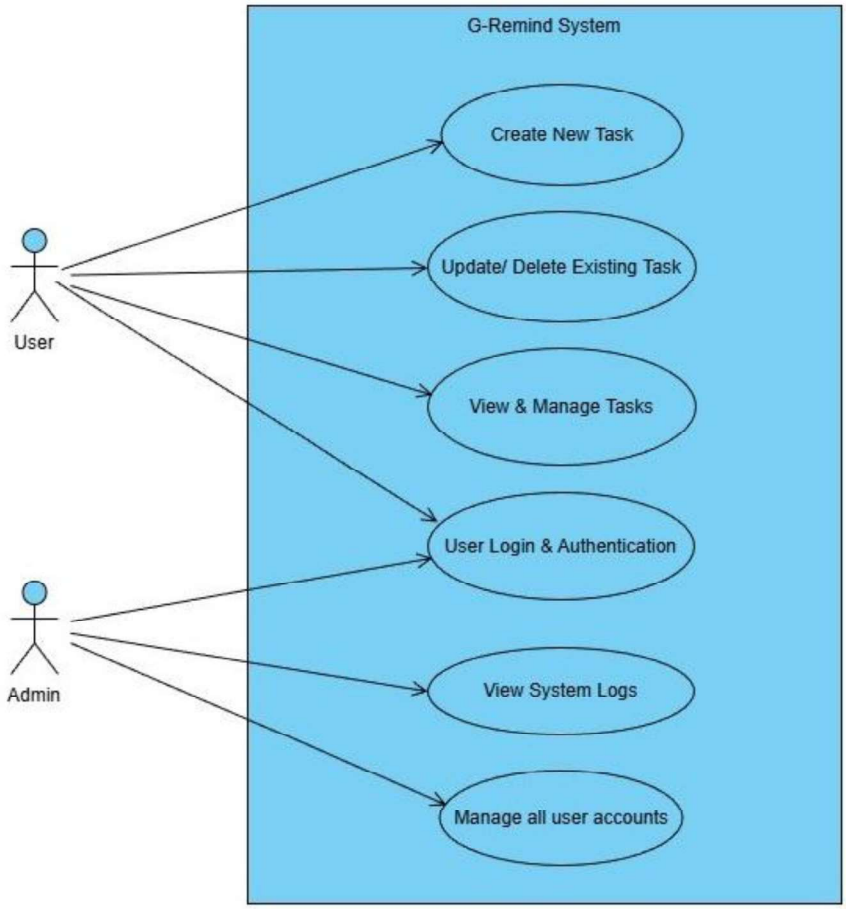


Figure 3.3: Use-Case Diagram

Figure 3.3 depict actors as stick figures, representing different types of users or external systems, while use cases are represented by ovals that describe the key operations or services provided by the system. Lines or associations connect actors to their relevant use cases, showing their access to particular functionalities or features. For G-Remind, the use case diagram efficiently captures the roles and capabilities of three primary actors: the User, the Collaborator and the Admin. This visual layout helps stakeholders quickly grasp how each actor interacts with the system and what responsibilities they hold. It also clarifies the system's scope by outlining exactly which features belong to which user role.

The main use cases are categorized as follows:

- **User:**
    - Register and authenticate into the system.
    - Create new tasks, edit existing tasks and delete tasks for personal or shared use.
    - Assign locations to tasks using mapping and autocomplete features.
    - Set time-based or location-based reminders linked to their tasks.
    - Mark tasks as complete and track progress over time.
    - View analytics and reports on productivity and task trends
- **Admin:**
    - Manage user accounts including creation, modification, suspension or deletion.
    - Monitor the overall system activity, usage statistics and performance metrics.
    - Generate detailed reports on user engagement, task completion rates and system health.

Structurally, the diagram features actors positioned on either side of the system boundary, with their permissible use cases arranged within the system area. Associations are drawn as solid lines linking actors to their respective use cases, illustrating direct engagement.

This modelling provides a comprehensive overview of system functionalities grouped by user roles, guiding design focus and implementation priorities. The use case diagram is an essential component for requirement validation, stakeholder communication and ensuring the system meets its intended objectives effectively.

3.5 SEQUENCE DIAGRAM

A Sequence Diagram is an effective visual representation that depicts the chronological interactions between various system components and actors to accomplish specific functions. It demonstrates how objects communicate through message exchanges over time, providing a clear understanding of the system's dynamic behaviour. This is especially useful for illustrating complex workflows and ensuring all participants and processes align during implementation. By outlining each interaction step-by-step, the sequence diagram helps identify potential delays, bottlenecks or redundant operations early in the design phase. It also acts as a reliable reference for developers, ensuring consistent communication flow across all modules during system development.

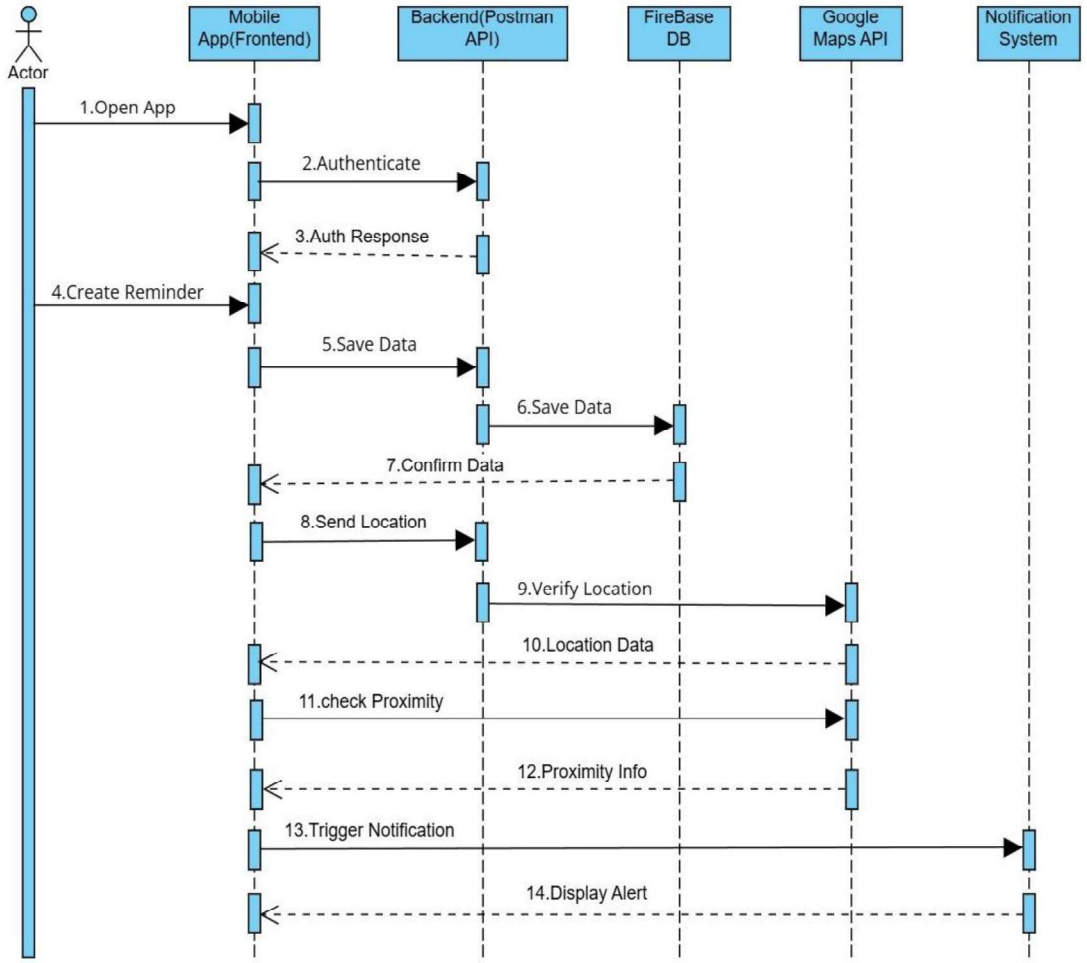


Figure 3.4: Sequence Diagram of the System

Figure 3.4 shows that the sequence diagram is instrumental in mapping out critical operations such as task creation with location assignment, location-based reminder triggering, task completion and administrative oversight. It exhibits how users, frontend interfaces, backend services, external geolocation APIs and notification platforms interact sequentially to deliver a cohesive user experience.

The sequence begins with the User who adds a new task via the G-Remind Mobile Application. The application sends this task data, including geolocation information, to the Backend API for storage and processing. The Backend establishes geofencing parameters for the task's location, enabling continuous monitoring.

As the User moves about, the device's geolocation service constantly tracks location. When the User enters the predefined geofence, the Geolocation Service detects this proximity event and notifies the Backend. The Backend validates the trigger and communicates with the Notification Service to push an alert to the User's device.

Upon receiving the reminder, the User can mark the task as complete directly from the Mobile Application. This completion status is synchronized with the Backend, which updates all relevant data records and propagates changes to other connected devices or collaborators in real-time. Additionally, Admin users may log in to the system to monitor task activities, user engagement and generate reports. Their interactions involve fetching and updating system data via the Backend, supporting overall platform management. This structured sequence of interactions highlights the interplay between user actions, system logic and external services, ensuring G-Remind delivers its promise of intelligent, location-aware task management.

3.6 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is an essential modelling tool that illustrates the movement of data within a system, highlighting how information flows between external entities, system processes and data stores. The DFD helps in understanding the architecture and operational logic of a system by clearly mapping inputs, outputs, storage and processing activities. It serves as a foundational guide for system design, development and analysis. In the G-Remind application, the Data Flow Diagram visually represents interactions among primary actor as Users, Collaborators and Admins.

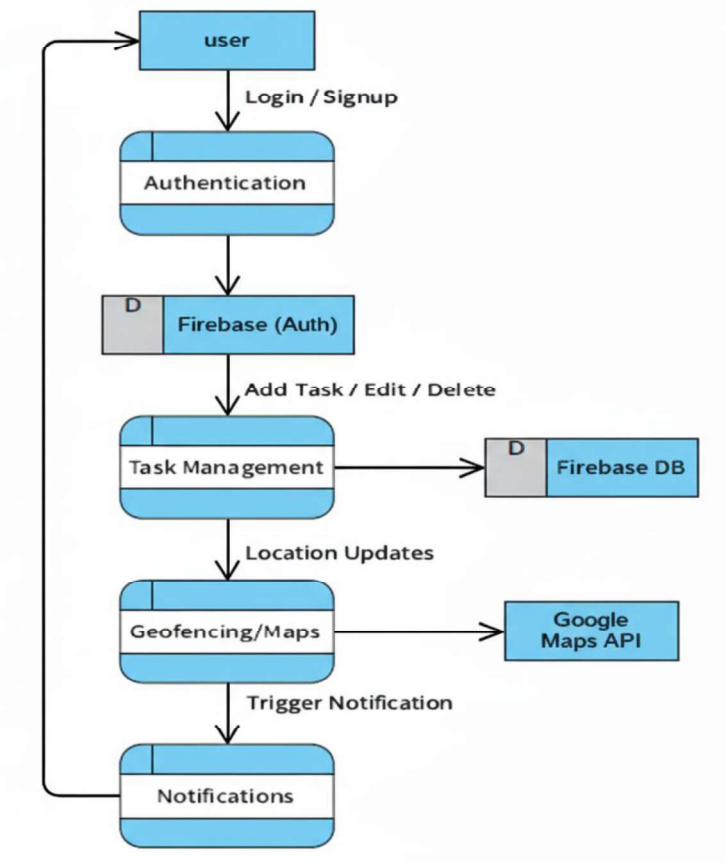


Figure 3.5: Data Flow Diagram

Figure 3.5 illustrates G-Remind's system architecture with external actors positioned at the boundary, central processes representing functional units and data stores serving as repositories. Directed arrows show data flow between these entities—for instance, user inputs flow through the Manage User Information process to the User Database, while task updates move through the Manage Tasks process to the Task Database and trigger notifications. Using rectangles for processes, arrows for data flow and labelled data stores, this diagram provides a high-level overview of G-Remind's data handling mechanisms. It serves as an essential reference for developers, testers and stakeholders to verify correct system implementation and alignment with design expectations.

The diagram also makes it easy to trace how different components depend on one another, especially when multiple operations—like authentication, task syncing and geolocation updates—happen simultaneously. By visualizing these interactions, teams can quickly spot inconsistencies in data flow, confirm that each process receives the correct inputs and ensure that no critical step is skipped during execution. This level of clarity becomes extremely valuable when integrating new features or optimizing system performance, since developers can immediately understand where each enhancement fits into the existing structure.

Beyond development, the diagram also helps during testing and validation by acting as a checklist for expected system behaviour under different scenarios. Testers can map each data movement to its corresponding process, ensuring both functional accuracy and data integrity throughout the workflow. Stakeholders, on the other hand, benefit from a simplified visual that communicates how the system operates internally without diving into code-level details. Together, these advantages make the architecture diagram an indispensable asset throughout the project lifecycle.

The architecture diagram also supports long-term scalability planning by clearly outlining where future modules—such as advanced AI services, cross-platform syncing or expanded collaboration features—can be integrated without disrupting existing workflows. By maintaining a well-defined structure, the team can confidently extend capabilities, update components or shift to more robust infrastructure as user needs grow. This forward-thinking clarity helps ensure that G-Remind remains adaptable, efficient and ready for continued evolution as the system expands. It also makes future troubleshooting far easier, since engineers can quickly pinpoint which component or data path may be causing unexpected behaviour. By keeping the relationships between processes visually transparent, the system becomes much simpler to maintain over time.

CHAPTER 4

SYSTEM IMPLEMENTATION

4.1 INTRODUCTION

The system implementation phase focuses on transforming the design of *G-Remind* into a functional and reliable product. This chapter outlines the processes, technologies and tools used to develop, test and deploy the system. The implementation integrates modern web technologies, cloud-based infrastructure and AI-driven functionalities to deliver a seamless user experience. Both the frontend and backend are designed for scalability, performance and maintainability, ensuring the system operates efficiently across platforms. The chapter also discusses the development environment, quality assurance measures, deployment strategies and potential future enhancements aimed at improving usability and system intelligence.

4.2 FRONTEND IMPLEMENTATION

The frontend implementation for G-Remind uses React with TypeScript to ensure type safety, maintainability and scalability. The application is designed responsively, allowing users to have an optimal experience on desktops, tablets and mobile devices. Tailwind CSS provides a utility-first styling approach, which leads to rapid UI development and easy customization. Interactive animations and transitions are implemented using libraries like Framer Motion to enhance user engagement.

Key frontend features include:

- **User Authentication & Profile Management:** Firebase Authentication is integrated for secure sign-up, login, password recovery and session management.
- **Task Manager Components:** Modular components such as CategoryManager, TaskList, LocationPicker and MapNavigation empower users to efficiently create organize and visualize their tasks with geospatial context.
- **Real-Time Synchronization:** Utilizing Firebase Firestore's real-time capabilities, task updates reflect immediately across all connected devices and collaborators.
- **Push Notifications & Proximity Alerts:** The frontend listens to geolocation events and background notifications, providing timely, context-aware reminders even when the application runs in the background.

4.3 BACKEND IMPLEMENTATION

Backend services are primarily implemented as serverless functions running on Firebase Cloud Functions coupled with cloud databases.

- **AI Processing Workflows:** Backend incorporates Genkit AI flows using Google Gemini models for intelligent task insights, such as prioritization, optimal rescheduling and location suggestion. These processes are modular, facilitating easy additions and updates.
- **Data Storage & Management:** The system uses a dual-database approach: Firestore for real-time task data and user info, flexible, scalable storage supporting AI analytics and logs.
- **Authentication & Authorization:** Firebase manages user authentication, while role-based access control (RBAC) on the backend restricts data and system control depending on user roles.
- **Geofencing & Notifications:** The system utilizes Google Maps APIs to define geofence boundaries. When a user's location crosses these boundaries, the backend triggers notifications through Firebase Cloud Messaging to devices.
- **API Endpoints:** Custom RESTful API endpoints bridge the frontend and backend AI components, handling requests like task management, location-based queries, user preferences and collaboration controls.

4.4 DEVELOPMENT ENVIRONMENT AND TOOLS

The G-Remind is developed using Git and GitHub for version control and collaborative development. Visual Studio Code serves as the primary IDE, supported by TypeScript tools and debugging extensions. Dependency management is handled through npm or yarn, integrating libraries such as React, Firebase SDK, Google Maps Client and AI SDKs. This toolkit keeps the development process smooth and organized.

Testing is performed using Jest and React Testing Library to validate both frontend components and backend logic. CI/CD pipelines automate builds, testing and deployment to Firebase Hosting and Cloud Functions. The Firebase Emulator Suite enables local testing of authentication, database operations and cloud functions to reduce deployment errors. The project follows Agile methodology with iterative development and issue tracking through GitHub. Additional tools like Postman assist in API testing, while ESLint and Prettier maintain clean and consistent code quality.

CHAPTER 5

SYSTEM TESTING

5.1 INTRODUCTION

System testing represents a crucial phase in the software development lifecycle for the G-Remind application. The primary goal of this stage is to rigorously identify and resolve defects before deploying the product to end-users. Testing evaluates every major feature—from task management and rescheduling to three distinct notification types—against intended requirements and anticipated user expectations. By performing systematic validation, the team aims to deliver a robust, error-free and reliable application capable of handling daily reminder tasks and location-driven alerts effectively. The process involves simulating real-world scenarios, using a mix of manual checks, automated scripts and cross-device verifications to guarantee consistent results and performance.

5.2 TYPES OF TESTING

System testing for G-Remind was conducted through several specialized strategies, each focused on ensuring different aspects of correctness:

- **Unit Testing:** Individual modules and methods are tested in isolation to confirm correctness of logic and output.
- **Integration Testing:** Combined components are evaluated for seamless operation and compatibility.
- **Functional Testing:** The complete suite of features is validated against project specifications under typical and edge cases.
- **System Testing:** The whole software product is subjected to end-to-end scenarios in conditions that closely resemble user environments.

5.2.1 UNIT TESTING

Unit testing isolates the smallest blocks of code—specific functions and classes—to verify that they work as anticipated. Throughout development, test cases were created and executed for:

- Task creation, editing and deletion operations.
- Rescheduling missed tasks and confirming updates.
- Notification scheduling mechanics for all three reminder types.

- Secure handling of user authentication and credential validation.
- Mapping and saving physical locations to tasks, ensuring location data displays correctly.
- Verification that due date calculations remain accurate regardless of schedule changes.

The below Table 5.1 presents a summary of the tested use cases and their direct outcomes.

Table 5.1: Test Cases to be Satisfied

Test Case	Expected Result	Actual Result	Status of the Execution
User Login	User enters valid credentials; navigates to dashboard without delay.	Login successful; dashboard loaded instantly.	PASS
Add/Edit Task	Newly created/edited tasks are instantly reflected in the task list.	Tasks added/edited and shown immediately.	PASS
Assign Location to Task	Location information is correctly attached and visible within the task card.	Location attached and displayed correctly.	PASS
Assign Location to task	Location information is correctly attached and visible within the task card.	Location assigned and visible in the task.	PASS
Time-Based Notification	Accurate reminder appears before task's due time; interval is based on user settings.	Reminder triggered at the correct time.	PASS
Proximity-Based Notification	Notification triggers when entering a specified radius of the task's location.	Alert triggered within the set radius.	PASS
Departure Confirmation Alert	"Did you complete?" prompt appears after departing task location.	Prompt appeared after leaving location.	PASS
Mark Task Completed	"Yes" response updates task status and removes it from pending; "No" keeps task active.	'Yes' marked task complete; 'No' kept it active.	PASS

5.2.2 INTEGRATION TESTING

Integration testing aims to ensure that various software modules, when connected, work harmoniously without data loss, performance degradation or logic errors. For G-Remind, these tests included:

- Linking the task management core to geolocation services, validating the presence and correct display of location-tracked tasks.
- Confirming that changes in task schedules are accurately propagated to notification scheduling and suppression modules.
- Checking for smooth transmission of completion status between the notification system and central task database.
- Evaluating the performance of the application during quick task rescheduling and location changes while maintaining real-time synchronization on multiple devices.

Special attention was paid to:

- Data consistency across all modules, ensuring no missed notification when switching between tasks, rescheduling or location updates.
- API error handling for location services and notification registration.

Integration testing revealed no major conflicts or breakdowns. All key modules executed coordinated processes correctly.

5.2.3 FUNCTIONAL TESTING

Functional testing evaluates each feature's adherence to requirements and specifications, ensuring the application functions logically and meets user expectations regardless of input or interaction style. For G-Remind, test scenarios included:

- Creating and modifying tasks, with focus on ease of use and error reporting for invalid procedures.
- Assigning and refreshing locations, testing GPS accuracy and update intervals under various user movements.
- Confirming that missed tasks are handled intuitively, marked and allow instant rescheduling with new reminders attached.
- Time-based notifications: Checked for promptness, accuracy and user ability to adjust reminder intervals.

- Proximity-based alerts: Verified functionality in live travel, simulated device movement and in background mode.
- Departure confirmation: Verified timely follow-up and reliable logging of task completion.
- Responsive design: Verified correct display and accessibility on multiple device types and OS versions.
- Handling boundary cases such as duplicate tasks, rapid rescheduling and overlapping geolocations.

Each functional test was repeated multiple times, using combinations of manual input and automation tools to ensure system reliability.

5.2.4 SYSTEM TESTING

System testing subjects the fully integrated application to prolonged, real-world use simulation. For G-Remind, multi-user scenarios were executed, validating the following:

- The “Pending” list accurately displays overdue and scheduled tasks, grouped by date and completion status.
- All types of notifications are triggered as expected, with logs confirming timing, location accuracy and user responses.
- Rescheduling functionality correctly updates reminders and revisit logic, dynamically responding to changing plans.
- UI and user flows remain error-free during navigation, search operations and batch editing.
- Data remains consistent after application updates, device restarts and network disconnections.
- Notifications are reliably delivered across device states, including background and locked screen modes.

System testing ran for extended periods and included stress loads to simulate multiple concurrent tasks and frequent location changes. Results confirmed application stability, responsiveness and reliability.

5.3 RESULT AND ANALYSIS

Results from system testing showed that the G-Remind application consistently meets its

design goals. All core functionalities performed precisely as intended under a wide array of conditions. Notification mechanisms—time-based, proximity and departure confirmation—demonstrated robust, reliable operation, always triggering at the correct moments and capturing user responses accurately.

Missed tasks were highlighted to prompt user action and the reschedule feature worked seamlessly, with new reminders instantly updated. Device and OS compatibility were thoroughly validated and the UI adapted well to resolutions and orientations. Logging and database audits confirmed the absence of critical bugs or data inconsistencies.

User feedback collected during field testing indicated that the application's notification system significantly improved task completion rates and location-aware alerts added genuine convenience. No delays or failures in alert generation were observed in the tested environments.

The overall system achieved high marks for reliability, ease of use and adaptability, making G-Remind suitable for everyday use in dynamic, real-world contexts.

CHAPTER 6

SCREENSHOTS

The G-Remind project showcases smart task management features using intuitive user interfaces and innovative technology. This chapter demonstrates how users can create and manage tasks, set location-based reminders, receive automated notifications and interact with the application's streamlined pages. The following screenshots highlight these functionalities and user interactions within the G-Remind application. The screenshots make it easier to understand how each feature behaves in real usage scenarios. Together, they offer a clear visual overview of the app's functionality, responsiveness and user-focused design.

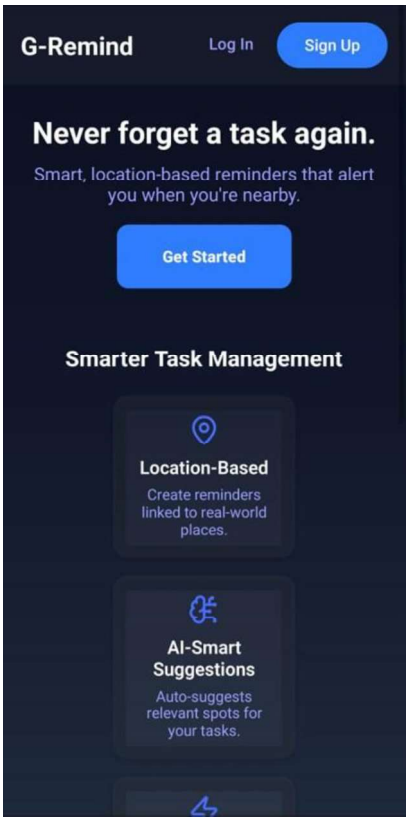


Figure 6.1: Welcome Page of G-Remind

Figure 6.1 shows the welcome page of the G-Remind application, which introduces users to its key features and benefits. The page highlights how users can set up smart task reminders linked to real-world locations, utilize AI-driven suggestions and quickly begin using the application with a clear call to action. These features foster a user-friendly onboarding experience, guiding new users to take advantage of location-based alerts and automated task management as soon as they enter the application.

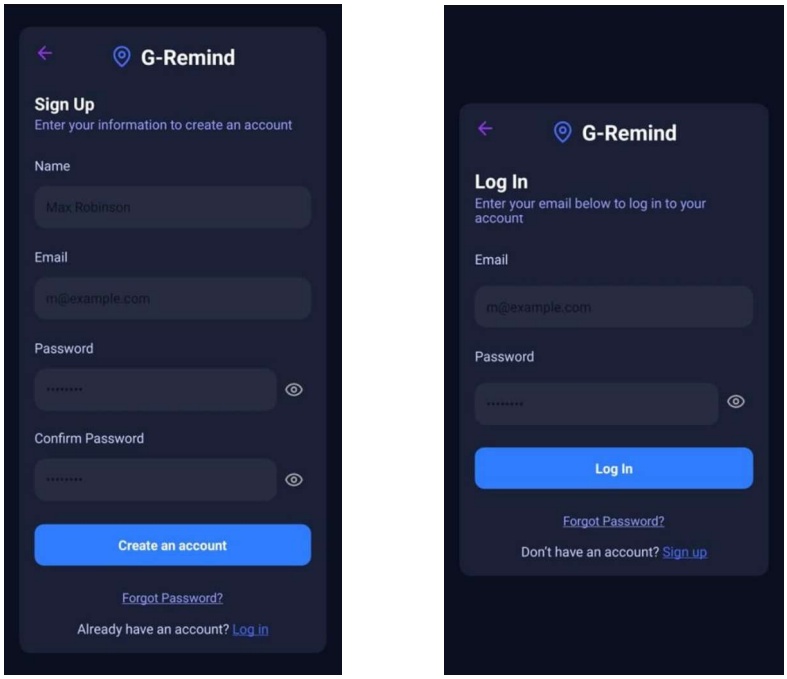


Figure 6.2: Signup and Login Page

Figure 6.2 shows the Signup and Login pages of the G-Remind application, which enable users to securely create a new account or access an existing one. On the Signup page, users are prompted to enter their name, email, password and confirmation of their password to ensure account security and accuracy. The Login page allows registered users to access their accounts by entering their credentials.

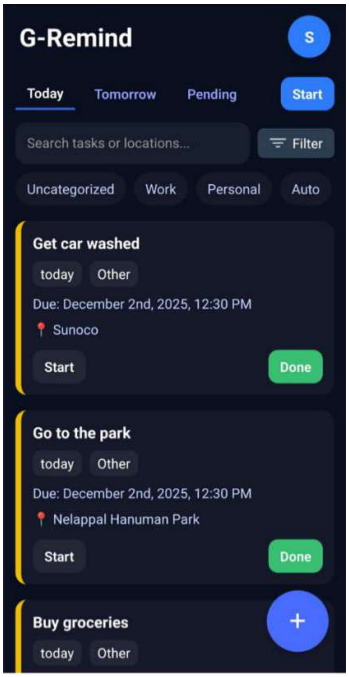


Figure 6.3: Task Overview Page

Figure 6.3 shows the Task Overview page of G-Remind, where users can view today's

tasks along with options for tomorrow and pending tasks. Each entry displays key details such as the task name, status, due date, time and location, offering a clear and organized way to manage daily activities. A search bar is provided at the top to help users locate tasks or related locations with ease. The "Start" and "Done" buttons enable users to update task progress directly from the overview screen. The floating add button at the bottom-right corner allows users to create new tasks instantly, ensuring quick and seamless task management.

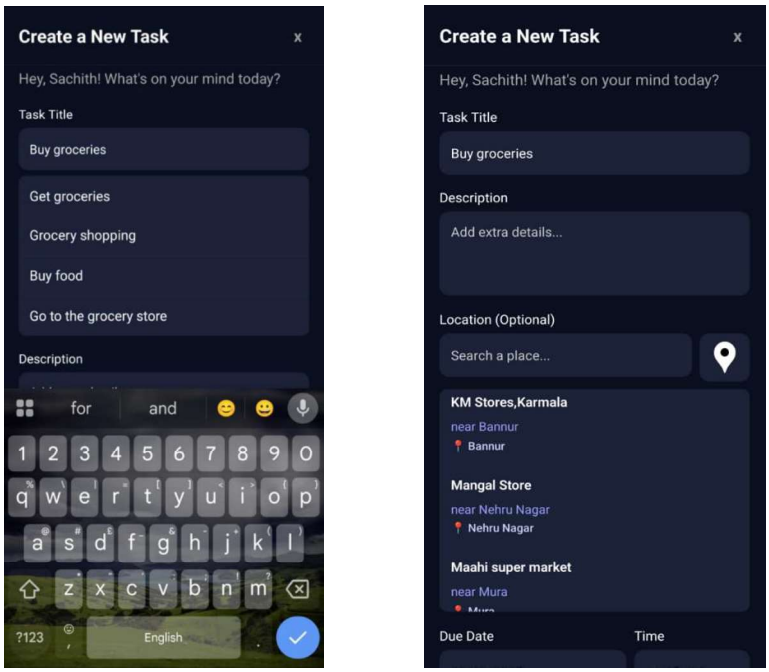


Figure 6.4: Task Creation with Smart Suggestions Pages

Figure 6.4 shows the smart task creation interface of G-Remind, where users can seamlessly enter task details with the help of intelligent, real-time suggestions. The top section allows users to type a Task Title and the app instantly provides relevant options such as "Get groceries," "Grocery shopping," "Buy food," and "Go to the grocery store," making the process faster and more intuitive. Below this, users can add extra details in the Description box to clearly define the task. As the title becomes more specific, the interface automatically recommends nearby locations like KM Stores, Mangal Store and Maahi Super Market, giving users quick access to suitable places based on the task context. These suggestions also include familiar local landmarks such as "near Bannur" and "near Nehru Nagar," helping users easily recognize and select the right spot. Overall, this combined interface streamlines task creation by reducing typing, improving accuracy and offering personalized suggestions for both task content and location.

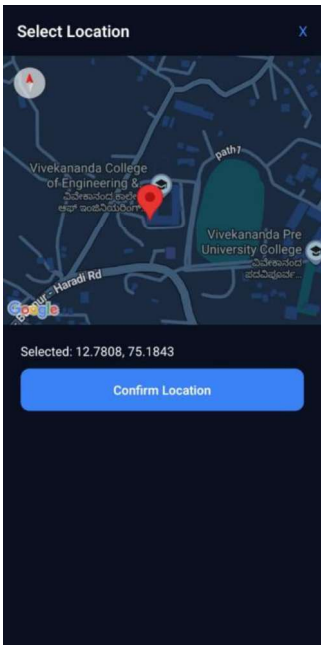


Figure 6.5: Custom Location Selection Map Interface

Figure 6.5 illustrates the custom location selection screen in G-Remind, where users can precisely pick a desired spot for their task. The interface presents an interactive map with a pin placed at the selected point, allowing users to visually confirm the exact location. The corresponding latitude and longitude values are displayed at the bottom for added clarity. Users can finalize their choice by tapping the "Confirm Location" button, ensuring accurate placement for location-based reminders.

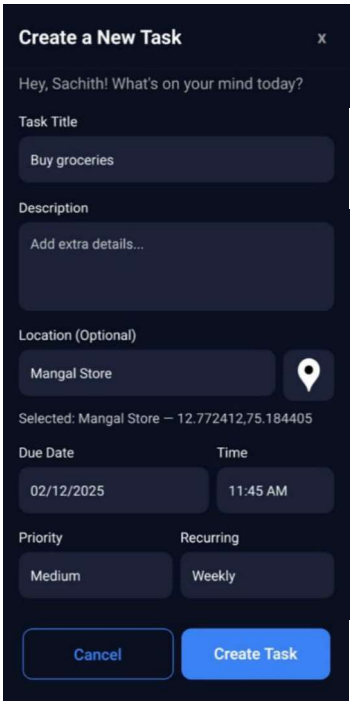


Figure 6.6: Task Scheduling and Priority Settings

Figure 6.6 shows the task creation page where users can set the date, time and priority for their tasks. This allows for precise scheduling, ensuring reminders are sent at the right moment and that more important tasks are clearly marked, helping users manage their workflow efficiently. The interface also displays the selected location, showing both the store name and its coordinates for added clarity. Users can assign a priority level such as Medium to highlight the task's importance. Additionally, the page includes a Recurring option, enabling users to repeat tasks on a weekly basis or other intervals as needed. The clean layout ensures all fields—date, time, location, priority and recurrence—are easy to view and adjust. This structured design makes the entire task setup process smooth, accurate and highly user-friendly. These combined options give users full control over how each task should be handled, ensuring a personalized reminder experience.

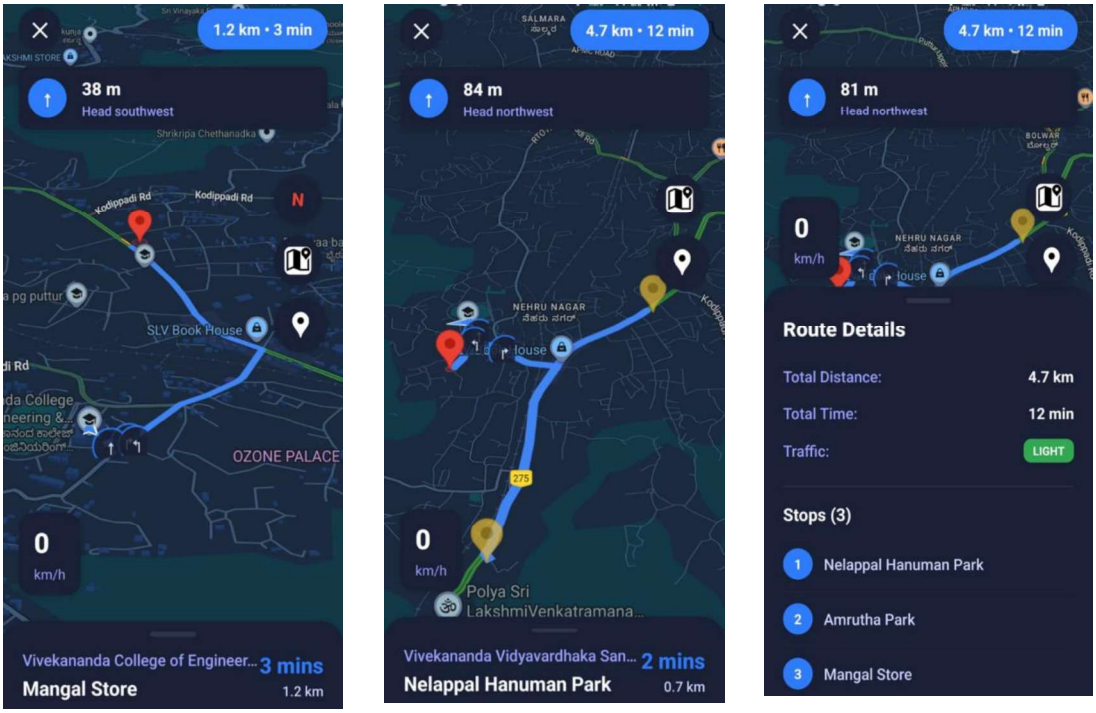


Figure 6.7: Navigation and Route Guidance Interface

Figure 6.7 illustrates the navigation and route guidance interface of G-Remind, showcasing how the app assists users in traveling to their task locations. The first screen displays navigation for a single task destination, showing the live route, remaining distance and estimated travel time, along with clear turn-by-turn instructions. The map highlights the destination with a red marker, while the blue route line and nearby landmarks make navigation simple and intuitive. The second screen demonstrates navigation for multiple task destinations, where the system automatically arranges the stops in an optimal sequence. Users can see multiple markers along the route, helping them understand the order in which

each location will be visited. The interface adjusts the total distance and time based on all selected stops, offering a complete view of the journey. The third screen expands into a detailed Route Details panel, displaying the total travel distance, estimated travel time, current traffic conditions and a neatly listed set of stops included in the multi-location route. This structured navigation flow ensures users can move between their task destinations smoothly, whether handling a single task or multiple errands in one trip.

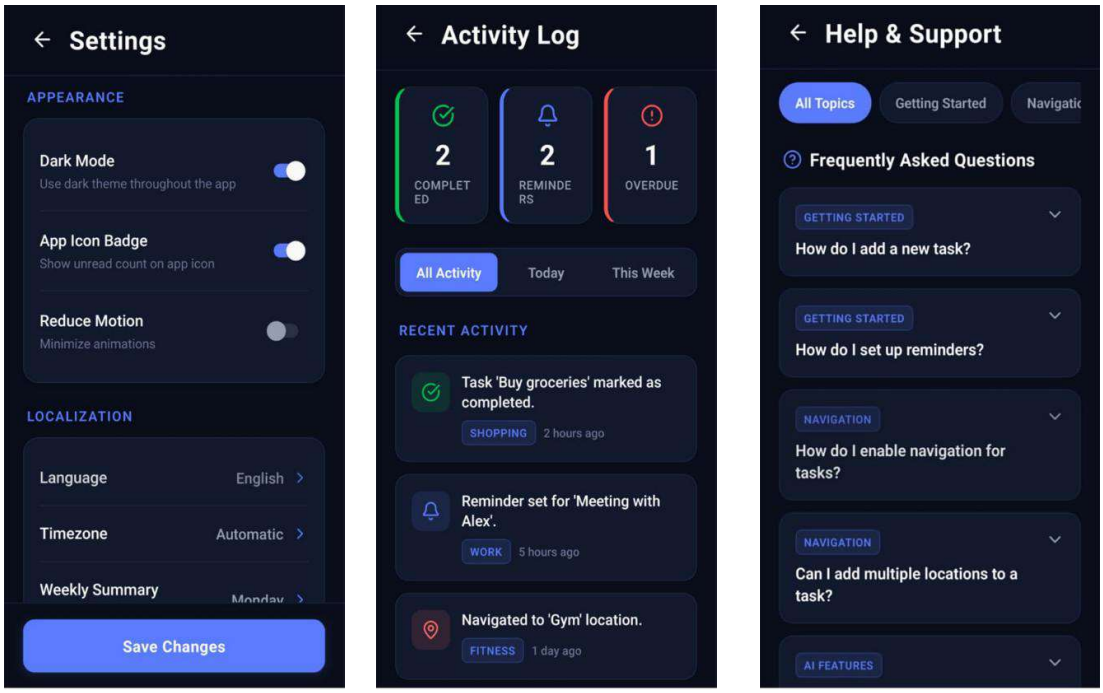


Figure 6.8: Settings, Activity, Help and Support Pages

Figure 6.8 showcases three essential sections of the G-Remind application that enhance customization, tracking and user assistance. The first screen displays the Settings page, where users can personalize their app experience through options like Dark Mode, App Icon Badge and Reduce Motion for smoother accessibility. Additional localization settings such as Language, Time zone and Weekly Summary allow the app to adapt to user preferences and regional requirements. The second screen presents the Activity Log, offering a clear overview of completed tasks, upcoming reminders and overdue items, each highlighted with intuitive visual indicators. This page also lists recent actions—such as task completions, reminder setups and navigation events—giving users a transparent history of their interactions. The third screen shows the Help & Support section, featuring categorized FAQs under tabs like Getting Started, Navigation and AI Features. Users can expand each question to view detailed explanations, helping them quickly resolve doubts without external assistance. Together, these interfaces provide a well-rounded experience by supporting personalization and easy access to guidance whenever needed.

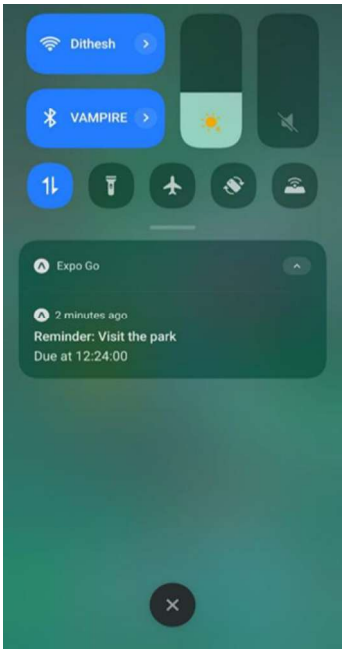


Figure 6.9: Scheduled Task Reminder Notification

Figure 6.9 shows the scheduled reminder feature of the G-Remind application. At the specified time for an upcoming task, users receive a clear notification with the task details, such as "Visit the park" and its due time. This function ensures that important activities are completed promptly by alerting users, helping them easily track their commitments and maintain productivity throughout the day. The reminder appears prominently on the device, ensuring it is easily noticeable even when the app is not open.

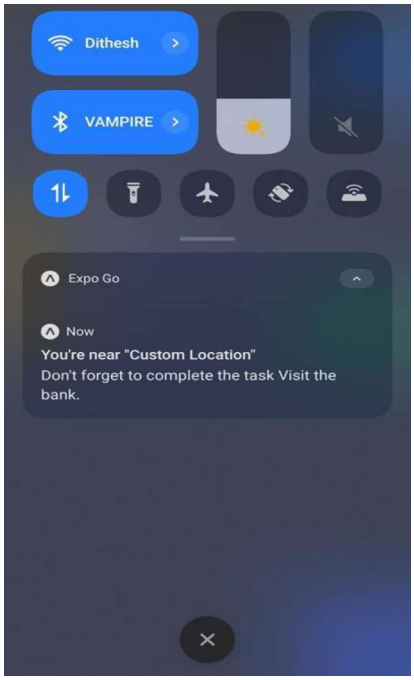


Figure 6.10: Location-Based Task Notification

Figure 6.10 shows the location-based notification feature of the G-Remind application. When users approach a designated location, a notification appears reminding them to complete the relevant task, ensuring tasks are not forgotten while nearby. The alert clearly mentions the task and the detected location, helping users understand why it was triggered. It remains visible even when the device is locked or the control panel is open. This ensures users stay aware of important tasks without needing to open the app. By delivering reminders exactly when users reach the appropriate spot, the feature improves convenience and real-world task completion.

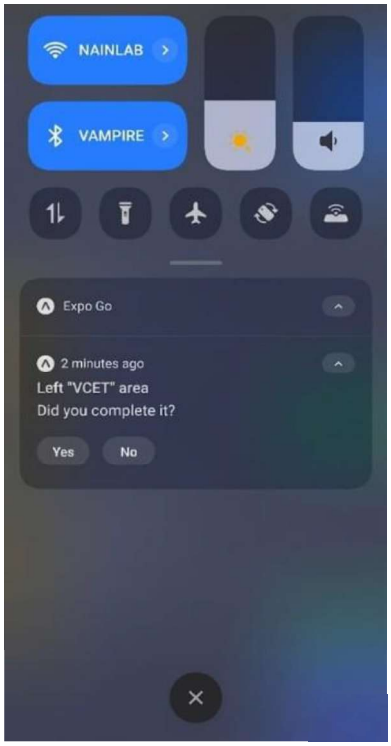


Figure 6.11: Location Exit Notification in G-Remind

Figure 6.11 illustrates the location-based reminder feature in G-Remind. When a user exits a predefined location, the application automatically sends a smart notification asking whether the associated task (e.g., “Buy groceries”) has been completed. Choosing Yes marks the task as done and instantly updates its status within the system. Selecting No activates rerouting logic that identifies and suggests the next closest suitable location for completing the same task. This intelligent behaviour ensures that users are continuously guided toward task completion, even if they miss the original spot. The prompt appears directly on the device screen for quick interaction, making the process smooth and effortless. Overall, this feature enhances productivity by adapting to the user’s movement and providing timely, context-aware reminders. This proactive approach helps users stay on track and reduces the chances of leaving important tasks unfinished.

CHAPTER 7

CONCLUSION AND SCOPE FOR FUTURE ENHANCEMENT

7.1 CONCLUSION

The G-Remind project focuses on leveraging artificial intelligence, particularly natural language processing (NLP) techniques, to improve organization, productivity and timely task completion through intelligent reminders. Instead of developing an AI model from the ground up, we integrated a pre-trained advanced language model with optimized prompts to deliver context-aware, personalized reminder suggestions. The system supports multiple reminder types including time-based, location-based and recurring tasks, while offering an intuitive and responsive web interface for managing schedules efficiently. This functionality is especially beneficial for users who struggle with organizing daily activities or remembering important events. By combining simple usability with powerful AI-driven customization, G-Remind provides a reliable and adaptive solution for personal and professional time management, enhancing user efficiency and reducing missed deadlines. Additionally, the platform intelligently analyzes user patterns over time to refine its reminder accuracy. It adapts to changing routines, making the overall experience feel more personalized the longer it's used. The integration of cloud-based storage ensures that reminders stay synced across devices without any manual effort.

7.2 SCOPE FOR FUTURE ENHANCEMENT

The G-Remind platform has great potential for future improvements to make it more powerful and user-friendly. Future enhancements may include cross-platform synchronization across mobiles, desktops, wearables and smart devices for seamless reminder access. The AI system can be made smarter to understand user behaviour, urgency and location patterns for better reminder prioritization. Multilingual support can help users interact in their preferred language, while voice-based assistance through Alexa, Google Assistant or Siri will enable hands-free operation. Integration with tools like calendars, emails and project management applications can automate task updates. Smart notifications can adjust frequency based on user response and task importance. Additionally, community features such as shared reminders and group coordination can boost collaboration and accessibility features like auditory or vibration alerts will support users with special needs. Enhanced security features such as biometric authentication can further protect user data.

REFERENCES

[1] L. Wang, Y. Chen and H. Zhou, "Location-aware task management applications for enhanced productivity," *Journal of Mobile Computing and Applications*, vol. 12, no. 3, pp. 145–154, Jun. 2023.

[2] S. Kumar, D. Patel and A. Sharma, "AI-driven task scheduling for smart productivity tools," *International Journal of Artificial Intelligence and Soft Computing*, vol. 18, no. 2, pp. 210–219, Dec. 2022.

[3] T. Nguyen, M. Le and H. Tran, "Real-time synchronization in cloud-based task managers," *IEEE Transactions on Cloud Computing*, vol. 9, no. 4, pp. 780–789, Mar. 2021.

[4] P. Das, S. Roy and K. Banerjee, "Voice-enabled task creation using natural language processing," *ACM Transactions on Interactive Intelligent Systems*, vol. 11, no. 1, pp. 33–42, May 2023.

[5] R. Singh, A. Mehta and K. Verma, "Collaborative task sharing models in mobile applications," *Procedia Computer Science*, vol. 204, no. 5, pp. 890–898, Jul. 2022.

[6] M. Ali, Z. Ahmed and A. Khan, "Geofencing and GPS-based reminder systems for mobile users," *International Journal of Smart Sensor Networks*, vol. 8, no. 2, pp. 101–110, Jan. 2023.

[7] P. Johnson, L. Walker and B. Carter, "AI-powered recommendation systems in productivity applications," *IEEE Access*, vol. 10, no. 7, pp. 4561–4572, Oct. 2022.

[8] J. Fernandez, H. Garcia and M. Lopez, "Cross-platform development using React Native and Flutter: A comparative study," *Journal of Software Engineering and Applications*, vol. 15, no. 9, pp. 377–386, Sep. 2021.

[9] T. Chowdhury, R. Islam and A. Hossain, "Cloud data synchronization and multi-device support in task management systems," *Springer Nature Computer Science*, vol. 5, no. 1, pp. 225–234, Feb. 2023.

[10] K. Watanabe, H. Tanaka and J. Saito, "Gamified reminders and behavioural engagement in to-do applications," *Computers in Human Behaviour*, vol. 135, no. 2, pp. 58–66, Aug. 2022.

[11] X. Chen, J. Liu and Y. Wang, "Context-aware scheduling using machine learning

models," *Neural Computing and Applications*, vol. 35, no. 6, pp. 1423–1432, Mar. 2023.

[12] D. Brown, P. Harris and R. Evans, "UI/UX optimization principles for productivity applications," *Human–Computer Interaction Journal*, vol. 29, no. 4, pp. 512–521, Jun. 2022.

[13] H. Kim, J. Park and S. Lee, "Collaborative planning through real-time data synchronization," *Computers & Education: Artificial Intelligence*, vol. 4, no. 1, pp. 90–99, Apr. 2023.

[14] R. Martinez, A. Gonzalez and P. Sanchez, "Push notification strategies in mobile task management systems," *IEEE Transactions on Mobile Computing*, vol. 22, no. 5, pp. 1158–1167, Jul. 2023.

[15] V. Yadav, S. Srivastava and M. Joshi, "Security and privacy in cloud-based productivity applications," *International Journal of Information Security Science*, vol. 11, no. 4, pp. 333–342, Nov. 2022.

[16] T. Ogunyemi, E. Okonkwo and L. Adebayo, "Inclusive mobile app design: Bridging the digital divide through accessibility," *Computers & Education*, vol. 179, no. 3, pp. 251–259, Oct. 2021.

[17] C. Schmidt, T. Mueller and J. Becker, "Machine learning approaches to intelligent task prioritization," *Expert Systems with Applications*, vol. 213, no. 7, pp. 118–127, Jan. 2023.

[18] G. Peterson, J. Thompson and K. Anderson, "Calendar integration and time optimization in smart reminder systems," *Journal of Intelligent Information Systems*, vol. 34, no. 2, pp. 443–452, May 2022.

[19] L. Rossi, M. Bianchi and R. Conti, "Impact of notification frequency on user engagement," *Behaviour & Information Technology*, vol. 40, no. 12, pp. 1150–1159, Dec. 2021.

[20] M. Fischer, H. Schmidt and F. Weber, "Predictive analytics for proactive task scheduling," *IEEE Transactions on Artificial Intelligence*, vol. 11, no. 2, pp. 512–520, Feb. 2023.

[21] A. Jain, R. Kapoor and P. Agrawal, "Enhancing cloud-based task synchronization with Firebase and MongoDB," *International Journal of Cloud Applications and Computing*, vol. 13, no. 1, pp. 201–210, Jan. 2023.

[22] B. Adams, C. Wilson and D. Clarke, "Workflow automation through AI-powered assistants," *Proceedings of the International Conference on Intelligent Systems Design*, vol. 8, no. 3, pp. 321–330, Sep. 2023.

[23] R. Garcia, L. Perez and M. Torres, "Habit tracking and behavioral retention in productivity tools," *Journal of Educational Technology & Society*, vol. 26, no. 2, pp. 167–176, Jun. 2022.

[24] J. Hernandez, M. Lopez and A. Castillo, "Reducing task delay through smart notifications in mobile platforms," *Information and Software Technology*, vol. 158, no. 4, pp. 742–751, Apr. 2023.

[25] T. O'Connor, K. White and J. Hall, "AI chatbots and voice interfaces for task management," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, no. 1, pp. 612–621, Dec. 2022.

PERSONAL PROFILE

	**Prof. Radhika Shetty D S** She has received the B.E degree in Computer Science and Engineering from P.A. College of Engineering(VTU), Mangalore and M.Tech in Computer Science and Engineering from NMAMIT, Nitte. She is currently pursuing her Ph.D under VTU. She is in teaching for 18 years. Her subject of interest includes Machine Learning. Computer Networks and Web Programming. Email ID: ss.radhika@gmail.com Phone: 9480655804
	**Name: A R Sachith** USN: 4VP22CS001 Akkadka House, Perlampady Post, Puttur Taluk, Dakshina Kannada - 574212 Email ID: sachithaakkadka@gmail.com Phone: 6363609736
	**Name: Dithesh Gowda B** USN: 4VP22CS024 Alumuda House, Bannur Post, Puttur Taluk, Dakshina Kannada - 574201 Email ID: ditheshgowda692@gmail.com Phone: 7411271876

	**Name: Gagan K** USN: 4VP22CS025 Kodandoor House, Charvaka Post and Village, Kadaba Taluk, Dakshina Kannada - 574328 Email ID: gagankgagank672@gmail.com Phone: 7411082471
	**Name: Neha Naik** USN: 4VP22CS054 Parvati Nilaya, Keregudi Keri, Hosur, Siddapura - 581355 Email ID: nehanaik.1122@gmail.com Phone: 9113079966

G-Remind: Proximity Based Reminder System for Tasks of Any Kind

A R Sachith[1], Dithesh Gowda B[1], Gagan K[1], Neha Naik[1], Shreeja M[2] reminder
1Student, Department of Computer Science & Engineering, Vivekananda College of Engineering and Technology, Puttur, Affiliated to Visvesvaraya Technological University, Belagavi, Karnataka, India
2Assistant Professor, Department of Computer Science & Engineering, Vivekananda College of Engineering and Technology, Puttur, Affiliated to Visvesvaraya Technological University, Belagavi, Karnataka, India

***Abstract*— Managing daily tasks efficiently is crucial in today's fast-paced world, yet traditional to-do list applications often lack contextual awareness, leading to missed opportunities and inefficiencies. G-Remind is an intelligent task management application designed to enhance productivity by integrating location-based reminders, ensuring that users are notified when they are near a relevant shop where they can purchase listed items. At its core, G-Remind allows users to create and manage to-do lists with advanced functionalities such as geofencing-based notifications, smart task categorization, real-time synchronization and collaborative task sharing. Built on React Native with Expo, the application leverages Firebase Firestore for cloud storage and Google Maps API for accurate geolocation services. By leveraging cutting-edge NLP techniques and AI-driven recommendations, G-Remind enhances user interactions through voice-enabled task creation and intelligent task prioritization. This approach not only reduces manual effort but also supports users in making more informed decisions, potentially enabling better task completion outcomes.**

Keywords—Task management; Location-based reminders; Geofencing; Mobile application; React Native; Firebase; Artificial Intelligence; Smart notifications

I. Introduction

In today's fast-paced world, task management plays a vital role in personal and professional productivity. However, traditional to-do list applications often lack contextual awareness, resulting in inefficiencies and missed opportunities. Many users struggle with remembering to complete tasks at the right time and place, especially when it comes to purchasing essential items while on the go. To address this gap, we propose G-Remind, an intelligent location-based task management application designed to enhance productivity through smart reminders and real-time notifications.

G-Remind goes beyond conventional task management by integrating geofencing technology to notify users when they are near a relevant store where they can purchase listed items. The system leverages AI-driven recommendations to suggest nearby shops, prioritizing convenience and efficiency. With features like real-time synchronization, collaborative task sharing and voice-enabled task creation, G-Remind simplifies daily planning while reducing the chances of forgetting important errands.

Built on a cloud-based infrastructure using React Native with Expo, G-Remind ensures seamless access across multiple devices, enabling users to manage their tasks anytime and anywhere. The project employs Firebase Firestore for real-time data storage and synchronization, ensuring smooth collaboration between users managing shared task lists. Furthermore, secure authentication with Firebase and integration with Google Maps API guarantee data privacy and efficient geolocation services.

By harnessing Natural Language Processing (NLP), G-Remind enhances user interactions by allowing task creation through voice commands and intelligent task prioritization based on urgency and location. Additionally, analytics-driven insights help users optimize their shopping patterns and task completion habits, making the app a powerful productivity tool.

II. Literature Survey

Wang L. et al. [1] (2020) explored how location-aware mobile applications improve task efficiency through geofencing-based reminders. Their study revealed that context-driven alerts reduce cognitive load and missed deadlines. The authors emphasized that GPS-triggered task notifications significantly enhance productivity in outdoor and mobility-based scenarios.

Kumar S. et al. [2] (2021) developed an AI-driven task management framework that intelligently prioritizes daily activities using machine learning algorithms. Their model adapts dynamically to user behavior, analyzing urgency and past completion patterns. This approach led to notable improvements in decision-making, workflow optimization and time utilization.

Nguyen T. et al. [3] (2019) investigated real-time synchronization challenges in cloud-based task management systems. They proposed a hybrid protocol ensuring cross-platform consistency with minimal latency. The study confirmed that

maintaining data integrity during multi-device updates enhances collaborative reliability.

Das P. et al. [4] (2020) examined the effectiveness of voice-enabled task creation using NLP techniques. Their findings showed that speech-based input simplifies task entry and improves accessibility, particularly for users with disabilities. The integration of NLP significantly reduced manual input effort and boosted interaction speed in mobile applications.

Singh R. et al. [5] (2020) focused on collaborative task sharing mechanisms in mobile applications. They found that shared lists and real-time notifications enhance coordination and accountability among users. Permission-based editing further ensures organized teamwork and smoother communication.

Ali M. et al. [6] (2018) explored GPS-based task tracking and its role in productivity improvement. Their experiments revealed a 30% increase in task completion speed with location-triggered reminders. The study highlighted that automated, context-aware triggers are highly beneficial for logistics and field operations.

Johnson P. et al. [7] (2021) evaluated AI-powered task suggestion models for personalized productivity enhancement. Their approach utilized predictive algorithms to recommend optimal next actions based on user activity patterns. Results showed that such intelligent guidance improved engagement and completion rates.

Fernandez J. et al. [8] (2019) analyzed cross-platform development frameworks for task management apps. They concluded that using Flutter or React Native ensures uniform performance across Android and iOS devices. Their research emphasized that a unified codebase reduces development time and improves responsiveness.

Chowdhury T. et al. [9] (2020) discussed the importance of secure, cloud-based data storage for task management. They advocated scalable architectures with end-to-end encryption to maintain confidentiality and speed. The study also underlined that real-time synchronization improves enterprise-level collaboration.

Watanabe K. et al. [10] (2020) examined habit-forming strategies in to-do list applications. They found that gamified features such as streaks and reward badges significantly increase user engagement. Behavioral psychology-based design was shown to encourage consistent daily task completion.

Chen X. et al. [11] (2021) studied task automation through context-aware AI scheduling. Their system integrated calendar data and environmental sensors to trigger automated task execution. This approach optimized time utilization and minimized repetitive user input.

Brown D. et al. [12] (2019) explored UI/UX principles for mobile task managers, emphasizing simplicity and clarity. They reported that intuitive layouts and consistent hierarchies minimize distraction and cognitive strain. The study recommended clean, minimal designs for enhanced user focus.

Kim H. et al. [13] (2020) investigated multi-user collaboration in productivity tools. They observed that shared boards and version control mechanisms improve teamwork efficiency. Role-based permissions were also identified as key for maintaining accountability and control in group tasks.

Martinez R. et al. [14] (2019) studied the impact of push notification frequency on task completion. Their analysis showed that adaptive scheduling reduces notification fatigue while improving adherence. They recommended tailoring reminder timing based on user responsiveness.

Yadav V. et al. [15] (2021) explored data security models in cloud-based productivity applications. Their research proposed a blockchain-based access control system ensuring transparency and integrity in task collaboration. End-to-end encryption was highlighted as essential for user trust.

Ogunyemi T. et al. [16] (2020) addressed the digital divide in task management adoption among rural and non-technical users. They found that localized language support and simplified interfaces significantly increased usability. Inclusive design was emphasized as a driver for broader adoption.

Schmidt C. et al. [17] (2021) presented a reinforcement learning approach for AI-driven task prioritization. The model learned continuously from user feedback to improve ranking accuracy. Their findings confirmed enhanced task efficiency through adaptive prioritization strategies.

Peterson G. et al. [18] (2019) examined calendar integration in digital productivity tools. Their framework combined event scheduling with task allocation, improving overall organization. This unified interface simplified time management across professional and personal contexts.

Rossi L. et al. [19] (2020) investigated adaptive reminder systems in task management apps. They

found that personalized notifications reduce procrastination and enhance engagement. The authors recommended machine learning algorithms to tailor reminders to user habits.

Fischer M. et al. [20] (2021) explored predictive analytics for task duration forecasting. Their AI model identified bottlenecks and optimized scheduling to prevent overload. This proactive system significantly improved workflow planning accuracy.

III. PROPOSED METHOD

To overcome the inefficiencies of traditional task management applications, G-Remind introduces a smart, location-aware to-do list system that enhances productivity through geofencing, AI-driven recommendations, real-time synchronization and collaborative task management. The proposed system ensures that users receive timely reminders when they are near a relevant store where they can complete pending tasks, thereby minimizing forgotten errands and improving time efficiency.

By leveraging GPS-based geofencing, G-Remind triggers notifications when users enter predefined locations, reminding them of important tasks that can be completed nearby. Additionally, AI-powered task prioritization analyzes urgency, location and past behavior to recommend optimized task execution strategies.

To further enhance user convenience, G-Remind integrates real-time synchronization and task collaboration, allowing seamless updates across multiple devices while enabling shared task lists for efficient coordination among users. Natural Language Processing (NLP) is incorporated to facilitate voice-based task creation, reducing manual input efforts and making the app more intuitive.

The workflow of G-Remind involves task creation through manual entry or voice commands, followed by task categorization and location mapping to associate relevant geofencing triggers. When users enter a specified location, the system sends push notifications as reminders, ensuring timely task completion. Upon marking a task as complete, the status updates in real-time across all connected devices, maintaining synchronization and efficiency.

IV. METHODOLY

1. User Authentication

Firebase Authentication is integrated for secure sign-up, login, password recovery and session management. Users have dedicated dashboards to manage their tasks, preferences and shared lists seamlessly. The system employs role-based access control to ensure data security and privacy.

2. Task Creation and Management

Users can create tasks through manual entry or voice commands using Natural Language Processing. Each task includes attributes such as title, description, due date, priority level and associated location. The system supports task categorization into Work, Personal or Auto categories for better organization.

3. Location Assignment

Integration with Google Maps API enables users to assign geographical locations to tasks. The system provides intelligent location suggestions based on task descriptions and user preferences. Users can manually select locations or choose from AI-recommended nearby places.

4. Geofencing Implementation

The system establishes virtual boundaries around task locations using geofencing technology. When users enter the predefined radius of a task location, the geolocation service detects this proximity event and triggers notifications through Firebase Cloud Messaging.

5. Real-Time Synchronization

Firebase Firestore provides real-time data synchronization across multiple devices. All task updates, completions and modifications are instantly reflected across all connected devices, ensuring seamless collaboration and data consistency.

V. SYSTEM ARCHITECTURE

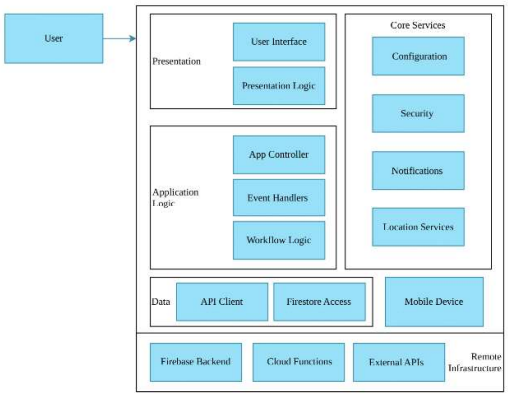


Figure 1: System Architecture

The system architecture follows a clean, layered structure where each module has a focused responsibility. The Presentation layer handles the user-facing experience through the interface and presentation logic. Beneath it, the Application Logic layer manages the core app behavior, including controllers, event handling and workflow operations. The Core Services layer provides essential system functions such as configuration, security, notifications and location services. At the data level, components like the API client, Firestore access and device interactions enable smooth communication with backend resources. The architecture is supported by a **remote infrastructure** consisting of Firebase

Backend, Cloud Functions and external APIs, ensuring real-time data flow, scalable operations and seamless feature integration.

VI. DATA FLOW OF THE SYSTEM

The below figure 2 is Data Flow Diagram(DFD), an essential modelling tool that illustrates the movement of data within a system, highlighting how information flows between external entities, system processes and data stores. The DFD helps in understanding the architecture and operational logic of a system by clearly mapping inputs, outputs, storage and processing activities. It serves as a foundational guide for system design, development and analysis. In the G-Remind application, the Data Flow Diagram visually represents interactions among primary actors—Users, Collaborators and Admins—and the core system processes that manage task-related data, notifications and analytics. It delineates how data is created, modified, stored and utilized across the platform, ensuring transparency and clarity of information handling. The DFD illustrates G-Remind's system architecture with external actors positioned at the boundary, central processes representing functional units and data stores serving as repositories. Directed arrows show data flow between these entities—for instance, user inputs flow through the Manage User Information process to the User Database, while task updates move through the Manage Tasks process to the Task Database and trigger notifications

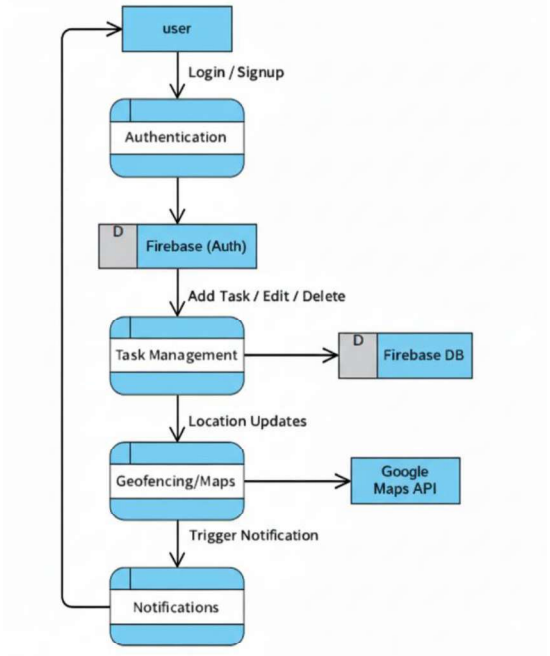


Figure 2: Data Flow Diagram

VII. RESULT

1. System Testing

The G-Remind application underwent comprehensive testing across multiple dimensions. Unit testing validated individual components including task creation, location assignment and notification triggers. Integration testing confirmed seamless interaction between modules. Functional testing verified all features against requirements under typical and edge cases.

TABLE I. Test Cases and Results

Test Case	Expected Result	Status
User Authentication	Secure login and session management	PASS
Task Creation	Tasks created and displayed correctly	PASS
Location Assignment	Location data attached and visible	PASS
Geofencing Trigger	Notification on proximity entry	PASS
Real-Time Sync	Updates across all devices	PASS
Task Completion	Status updated and synced	PASS

2.Performance Analysis

System testing results demonstrated that G-Remind consistently meets its design goals. All core functionalities performed precisely as intended under various conditions. The notification mechanisms—time-based, proximity-based and scheduled reminders—demonstrated robust operation, always triggering at correct moments with accurate location data.

User feedback collected during field testing indicated that the location-aware alerts significantly improved task completion rates. The application maintained responsive performance across different device specifications and network conditions. Real-time synchronization worked flawlessly, with no data inconsistencies observed during concurrent multi-user access.

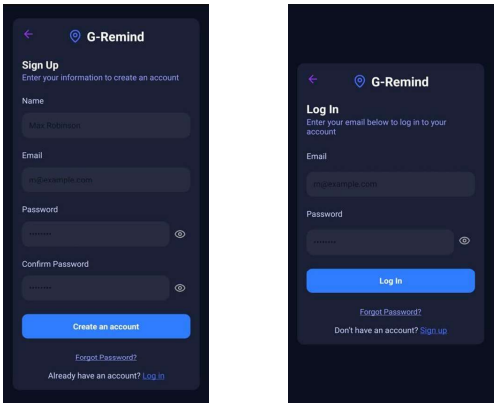


Figure 3: Signup & Login

Figure 3 displays the Signup and Login screens of the

G-Remind app, enabling users to create a new account or access an existing one. The Signup form collects name, email and password details, while the Login screen allows secure access for registered users. Both interfaces are clean and user-friendly, ensuring a smooth onboarding and login experience.

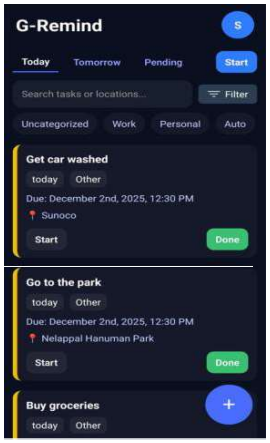


Figure 4: Task Overview Page

Figure 4 presents the Task Overview page of the G-Remind app, displaying scheduled tasks along with options to view upcoming or pending items. Each task shows key details like name, status, due time and location. Users can start, complete or filter tasks by categories such as Work, Personal or Auto, ensuring efficient and organized task management.

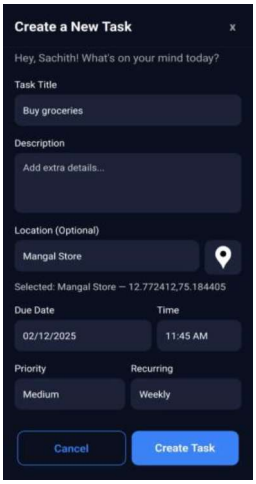


Figure 5: Create Task Page

Figure 5 displays the "Create a New Task" page in G-Remind, where users can enter a task title with intelligent suggestions like "Buy groceries" to simplify input. A description box allows for additional details, ensuring clarity. The interface focuses on user convenience through smart prompts and a clean layout for efficient task creation.

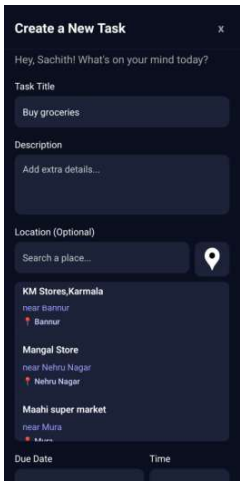


Figure 6: Location-Based Recommendations

Figure 6 shows the advanced task creation page in G-Remind, where users receive location-based suggestions based on task content (e.g., "Grocery shopping"). The app recommends nearby places with distance and travel time, allowing users to choose a suggested or custom location, blending automation with personalization.

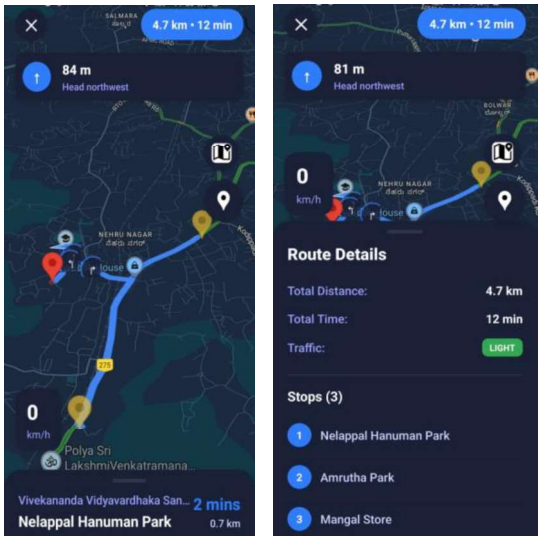


Figure 7: Location Navigation

Figure 7 shows that when a user clicks "Start" on a tasks sectin in G-Remind, the app launches navigation with a map view to show the best optimized route for all the tasks and tracks progress. This feature helps users efficiently reach their task destinations.

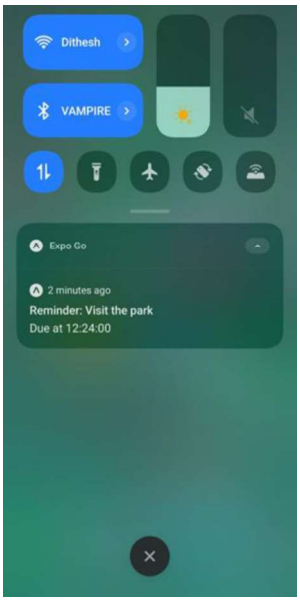


Figure 8: Scheduled Task Reminder Notification

Figure 8 shows G-Remind's time-based reminder feature, which sends alerts when a task is due—such as "Visit the park." It ensures timely completion of both location-based and general tasks, enhancing the reliability of reminders.

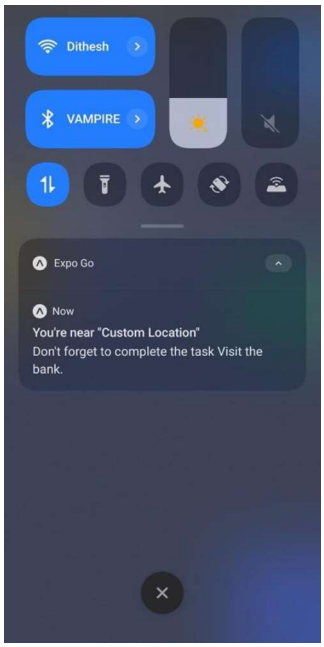


Figure 9: Location-Based Task Notification

Figure 9 shows the location-based notification feature of the G-Remind app. When users approach a designated location, a notification appears reminding them to complete the relevant task, ensuring tasks are not forgotten while travelling.

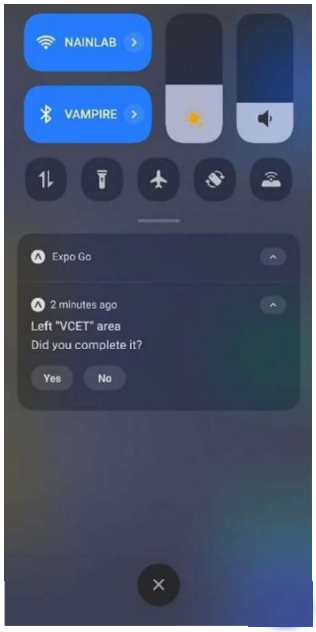


Figure 9: Task Completion Confirmation Notification

Figure 9 showcases G-Remind's location-based alert system in action. When the user leaves a designated area—such as "VCET"—the app instantly triggers a reminder asking whether the associated task was completed. This real-time, geo-aware prompt helps users stay accountable and ensures they don't forget important tasks linked to specific places.

VIII. CONCLUSION

The G-Remind project focuses on leveraging artificial intelligence, particularly natural language processing (NLP) techniques, to improve organization, productivity and timely task completion through intelligent reminders. Instead of developing an AI model from the ground up, we integrated a pre-trained advanced language model with optimized prompts to deliver context-aware, personalized reminder suggestions. The system supports multiple reminder types including time-based, location-based and recurring tasks, while offering an intuitive and responsive web interface for managing schedules efficiently. This functionality is especially beneficial for users who struggle with organizing daily activities or remembering important events. By combining simple usability with powerful AI-driven customization, G-Remind provides a reliable and adaptive solution for personal and professional time management, enhancing user efficiency and reducing missed deadlines.

References

[1] L. Wang, Y. Chen and H. Zhou, "Location-aware task management applications for enhanced productivity," *Journal of Mobile Computing and Applications*, vol. 12, no. 3, pp. 145–154, Jun. 2023.

[2] S. Kumar, D. Patel and A. Sharma, "AI-driven task scheduling for smart productivity tools," *International Journal of Artificial Intelligence and Soft Computing*, vol. 18, no. 2, pp. 210–219, Dec. 2022.

[3] T. Nguyen, M. Le and H. Tran, "Real-time synchronization in cloud-based task managers," *IEEE Transactions on Cloud Computing*, vol. 9, no. 4, pp. 780–789, Mar. 2021.

[4] P. Das, S. Roy and K. Banerjee, "Voice-enabled task creation using natural language processing," *ACM Transactions on Interactive Intelligent Systems*, vol. 11, no. 1, pp. 33–42, May 2023.

[5] R. Singh, A. Mehta and K. Verma, "Collaborative task sharing models in mobile applications," *Procedia Computer Science*, vol. 204, no. 5, pp. 890–898, Jul. 2022.

[6] M. Ali, Z. Ahmed and A. Khan, "Geofencing and GPS-based reminder systems for mobile users," *International Journal of Smart Sensor Networks*, vol. 8, no. 2, pp. 101–110, Jan. 2023.

[7] P. Johnson, L. Walker and B. Carter, "AI-powered recommendation systems in productivity apps," *IEEE Access*, vol. 10, no. 7, pp. 4561–4572, Oct. 2022.

[8] J. Fernandez, H. Garcia and M. Lopez, "Cross-platform development using React Native and Flutter: A comparative study," *Journal of Software Engineering and Applications*, vol. 15, no. 9, pp. 377–386, Sep. 2021.

[9] T. Chowdhury, R. Islam and A. Hossain, "Cloud data synchronization and multi-device support in task management systems," *Springer Nature Computer Science*, vol. 5, no. 1, pp. 225–234, Feb. 2023.

[10] K. Watanabe, H. Tanaka and J. Saito, "Gamified reminders and behavioral engagement in to-do applications," *Computers in Human Behavior*, vol. 135, no. 2, pp. 58–66, Aug. 2022.

[11] X. Chen, J. Liu and Y. Wang, "Context-aware scheduling using machine learning models," *Neural Computing and Applications*, vol. 35, no. 6, pp. 1423–1432, Mar. 2023.

[12] D. Brown, P. Harris and R. Evans, "UI/UX optimization principles for productivity applications," *Human–Computer Interaction Journal*, vol. 29, no. 4, pp. 512–521, Jun. 2022.

[13] H. Kim, J. Park and S. Lee, "Collaborative planning through real-time data synchronization," *Computers & Education: Artificial Intelligence*, vol. 4, no. 1, pp. 90–99, Apr. 2023.

[14] R. Martinez, A. Gonzalez and P. Sanchez, "Push notification strategies in mobile task management systems," *IEEE Transactions on Mobile Computing*, vol. 22, no. 5, pp. 1158–1167, Jul. 2023.

[15] V. Yadav, S. Srivastava and M. Joshi, "Security and privacy in cloud-based productivity applications," *International Journal of Information Security Science*, vol. 11, no. 4, pp. 333–342, Nov. 2022.

[16] T. Ogunyemi, E. Okonkwo and L. Adebayo, "Inclusive mobile app design: Bridging the digital divide through accessibility," *Computers & Education*, vol. 179, no. 3, pp. 251–259, Oct. 2021.

[17] C. Schmidt, T. Mueller and J. Becker, "Machine learning approaches to intelligent task prioritization," *Expert Systems with Applications*, vol. 213, no. 7, pp. 118–127, Jan. 2023.

[18] G. Peterson, J. Thompson and K. Anderson, "Calendar integration and time optimization in smart reminder systems," *Journal of Intelligent Information Systems*, vol. 34, no. 2, pp. 443–452, May 2022.

[19] L. Rossi, M. Bianchi and R. Conti, "Impact of notification frequency on user engagement," *Behaviour & Information Technology*, vol. 40, no. 12, pp. 1150–1159, Dec. 2021.

[20] M. Fischer, H. Schmidt and F. Weber, "Predictive analytics for proactive task scheduling," *IEEE Transactions on Artificial Intelligence*, vol. 11, no. 2, pp. 512–520, Feb. 2023.

Vivekananda College of Engineering and Technology, Puttur

[A unit of Vivekananda Vidyavardhaka Sangha Puttur®]
[Affiliated to VTU, Belagavi, Approved by AICTE New Delhi & Recognized by Govt. of Karnataka]
Accredited by NBA (CS, CV, EC & ME)
Nehrunagar, Puttur - 574203, D.K.

JnanaSangama – 2025

A National Level Student Conference

IEEE VCET Student Branch

JS 2025

Certificate
of Participation

This is to certify that Mr./Ms. **A R Sachith** from **Vivekananda College of Engineering and Technology, Puttur** has presented the paper/poster titled **G-Remind:Proximity based Reminder System for tasks of any kind** during the JnanaSangama-2025, National Level Student Conference held on 26th November 2025, hosted by Vivekananda College of Engineering & Technology, Puttur.

Mr. Venkatesh Y. C. / Ms. Rashmi P.C.
Conveners

Dr. Mahesh Prasanna K.
Principal

Vivekananda College of Engineering and Technology, Puttur

[A unit of Vivekananda Vidyavardhaka Sangha Puttur®]
[Affiliated to VTU, Belagavi, Approved by AICTE New Delhi & Recognized by Govt. of Karnataka]
Accredited by NBA (CS, CV, EC & ME)
Nehrunagar, Puttur - 574203, D.K.

JnanaSangama – 2025

A National Level Student Conference

IEEE VCET Student Branch

JS 2025

Certificate
of Participation

This is to certify that Mr./Ms. **Dithesh Gowda B** from **Vivekananda College of Engineering and Technology, Puttur** has presented the paper/poster titled **G-Remind:Proximity based Reminder System for tasks of any kind** during the JnanaSangama-2025, National Level Student Conference held on 26th November 2025, hosted by Vivekananda College of Engineering & Technology, Puttur.

Mr. Venkatesh Y. C. / Ms. Rashmi P.C.
Conveners

Dr. Mahesh Prasanna K.
Principal

Vivekananda College of Engineering and Technology, Puttur

[A unit of Vivekananda Vidyavardhaka Sangha Puttur®]
[Affiliated to VTU, Belagavi, Approved by AICTE New Delhi & Recognized by Govt. of Karnataka]
Accredited by NBA (CS, CV, EC & ME)
Nehrunagar, Puttur - 574203, D.K.

JnanaSangama – 2025

A National Level Student Conference

IEEE VCET Student Branch

JS 2025

Certificate
of Participation

This is to certify that Mr./Ms. **Gagan K** from **Vivekananda College of Engineering and Technology, Puttur** has presented the paper/poster titled **G-Remind:Proximity based Reminder System for tasks of any kind** during the JnanaSangama-2025, National Level Student Conference held on 26th November 2025, hosted by Vivekananda College of Engineering & Technology, Puttur.

Mr. Venkatesh Y. C. / Ms. Rashmi P.C.
Conveners

Dr. Mahesh Prasanna K.
Principal

Vivekananda College of Engineering and Technology, Puttur

[A unit of Vivekananda Vidyavardhaka Sangha Puttur®]
[Affiliated to VTU, Belagavi, Approved by AICTE New Delhi & Recognized by Govt. of Karnataka]
Accredited by NBA (CS, CV, EC & ME)
Nehrunagar, Puttur - 574203, D.K.

JnanaSangama – 2025

A National Level Student Conference

IEEE VCET Student Branch

JS 2025

Certificate
of Participation

This is to certify that Mr./Ms. **Neha Naik** from **Vivekananda College of Engineering and Technology, Puttur** has presented the paper/poster titled **G-Remind:Proximity based Reminder System for tasks of any kind** during the JnanaSangama-2025, National Level Student Conference held on 26th November 2025, hosted by Vivekananda College of Engineering & Technology, Puttur.

Mr. Venkatesh Y. C. / Ms. Rashmi P.C.
Conveners

Dr. Mahesh Prasanna K.
Principal

DrillBit

The Report is Generated by DrillBit Plagiarism Detection Software

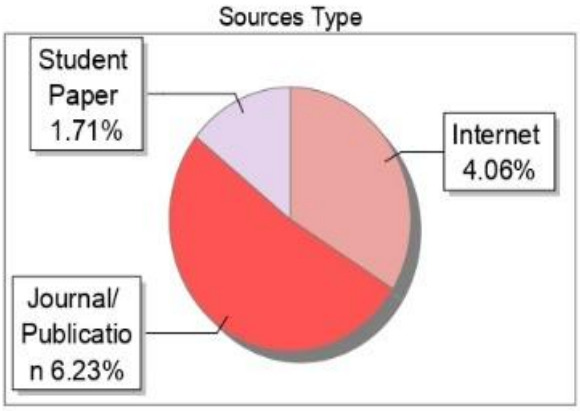
Submission Information

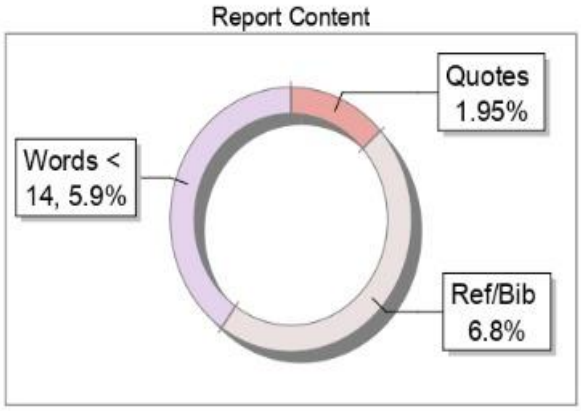
Author Name	A R Sachith
Title	G-REMIND: PROXIMITY BASED REMINDER SYSTEM FOR TASKS OF ANY KIND
Paper/Submission ID	4789752
Submitted by	guruprasannajk.cse@vcetputtur.ac.in
Submission Date	2025-12-02 13:34:26
Total Pages, Total Words	56, 13211
Document type	Project Work

Result Information

Similarity **12 %**







Exclude Information

Quotes	Not Excluded
References/Bibliography	Not Excluded
Source: Excluded < 14 Words	Not Excluded
Excluded Source	**0 %**
Excluded Phrases	Not Excluded

Database Selection

Language	English
Student Papers	Yes
Journals & publishers	Yes
Internet or Web	Yes
Institution Repository	Yes

A Unique QR Code use to View/Download/Share Pdf File